

Working with nflverse Data in Python

Every NFL play since 1999 is a free download. Learn to load it, read it, filter it honestly, join the charting data to it, and rebuild this book's charts yourself.

Super Bowl LX, early in the second quarter.¹ Seattle has second-and-10 at its own 24, and Kenneth Walker III takes the handoff around the left end for 30 yards before he is pushed out of bounds at the New England 46. On the broadcast it is a highlight. By the next morning it is also a line of data: (14:04) (Shotgun) 9-K.Walker left end pushed ob at NE 46 for 30 yards (25-M.Jones) ., plus 371 other columns that record the down, the distance, the yard line, the expected points before and after, Seattle's chance of winning, who carried and who made the tackle. Every chart in this book was built from lines like that one. This chapter teaches you to load them, read them, filter them, join them to the charting data that says who was on the field, and rebuild the charts you have already seen. Along the way you'll find out why two respectable sources disagree about whether Seattle's quarterback was even in the shotgun on that play.

You'll be able to:

- Install nflreadpy, load a season of play-by-play in two lines, and move between Polars and pandas when you need to.
- Load schedules, rosters, participation and FTN charting data, and say what each one adds and which seasons it covers.
- Read a play-by-play row: find the situation (`down`, `ydstogo`, `yardline_100`), the result, and the model columns (`ep`, `epa`, `wp`).
- Filter to “real” plays (runs and dropbacks, without plays wiped out by penalties, kneel-downs or garbage time) and state exactly which filter you used.
- Join participation and FTN charting data to play-by-play for personnel and formation, without being fooled by the 2022-to-2023 source break.
- Reproduce a chart from an earlier chapter end to end, and score your prediction log against the new data every week.

You'll need

- **Downs, Distance, and Field Position:** `down` and `distance`, the yard line, **expected points** (what a situation is worth on average), **EPA** (how much one play changed it) and **success rate** (the share of plays with positive EPA). This chapter shows where those numbers live in the data.
- **Personnel: Who's on the Field Before Anything Else:** **personnel groupings**, the two-digit code for backs and tight ends (**11 personnel** is one back, one tight end, three receivers). We rebuild that chapter's season-by-season chart.
- **Python:** you should be comfortable with pandas (`groupby`, `merge`, boolean filters). Nothing else is assumed.

How to read this chapter

Part 13 shows its code; the other parts fold it away. Read the cells as part of the text, and if you can, run them as you go: each cell builds on the ones above it, top to bottom, the way a notebook does. The first cell imports what the chapter needs and defines one small helper that prints a data frame as a table. The football starts in the next section.

```
import re

import numpy as np
import pandas as pd
import polars as pl
import matplotlib.pyplot as plt
from matplotlib.colors import LinearSegmentedColormap
from matplotlib.patches import Circle, FancyArrowPatch, FancyBboxPatch, Rectangle
from IPython.display import Markdown, display
```

¹Super Bowl LX, February 8, 2026, Levi's Stadium: Seattle 29, New England 13; Kenneth Walker III was the game's MVP with 27 carries for 135 yards (https://en.wikipedia.org/wiki/Super_Bowl_LX; course fact sheet). The play is nflverse game 2025_22_SEA_NE, play 1015.

```

from gridiron import Play, Field, offense, defense, style, show_animation
from gridiron.data import nfl, pbp

style.apply()

def drop_numbers(ax):
    """Remove the painted yard numbers from a gridiron field (they sit under wide players)."""
    for txt in list(ax.texts):
        if txt.get_text().strip().isdigit():
            txt.remove()

def md_table(df, digits=2):
    """Show a small DataFrame as a Markdown table (presentation plumbing, not football)."""
    out = df.copy()
    for c in out.select_dtypes("number"):
        whole = (out[c].dropna() % 1 == 0).all() # downs, yard lines, counts
        sep = "" if c in ("season", "week", "play_id") else "," # 2025, not 2,025
        out[c] = out[c].map(lambda v: f"{v:{sep}.{0 if whole else digits}f}")
    align = ["---:" if pd.api.types.is_numeric_dtype(df[c]) else ":-:" for c in df]
    rows = [list(out.columns), align] + out.astype(str).values.tolist()
    return Markdown("\n".join("| " + " | ".join(map(str, r)) + " |" for r in rows))

```

nflverse: free football data, and why it exists

nflverse is a volunteer, open-source project that publishes free, cleaned NFL data and the packages that load it. Its centrepiece is **play-by-play** data, usually shortened to **pbp**: one row for every play of every game since the 1999 season, about 48,000 rows a season, each with a few hundred columns.

Where does it come from? The raw material is the league's own record of each game: the play descriptions typed up by the official statisticians in the press box, the same text that fills the gamebook a reporter picks up after the game. That record says what happened in words and a handful of fields. nflverse turns it into a table, fixes its inconsistencies (team abbreviations that changed when franchises moved, plays logged out of order, penalties enforced from odd spots) and then adds the columns that make it useful for analysis: who was the passer and the receiver, expected points and EPA, win probability, completion probability and more. Those model columns are the reason the data matters. Anyone can count yards; EPA on every play since 1999, free, is what let public analysts argue with professional ones.

The project has a short, busy history. It began at Carnegie Mellon, where the statisticians Maksim Horowitz, Ronald Yurko and Samuel Ventura built **nflscrapR**, an R package that scraped the league's game data and fitted the first widely used public expected-points and win-probability models.² When the league changed its data feeds around 2020, Ben Baldwin and Sebastian Carl wrote **nflfastR**, a faster rebuild with new models, and the family of packages around it took the name nflverse.³ None of these people worked for a team when they started. Horowitz began nflscrapR as an undergraduate; Baldwin was writing about football analytics in his spare time. The data exists because a handful of fans with statistics training decided it should be free.

For Python users, the loader is **nlfireadpy**: one function per table, each returning a Polars data frame. It replaced **nfl_data_py**, the older Python package, which is now archived; its own README says it is deprecated in favour of nlfireadpy and will get no further maintenance.⁴ In October 2026 the current release of nlfireadpy is 0.1.5 (November 2025), and the repository is still active.⁵

²Maksim Horowitz, Ronald Yurko and Samuel Ventura, nflscrapR (R package): <https://github.com/maksimhorowitz/nflscrapR>; the expected-points and win-probability models are described in Ronald Yurko, Samuel Ventura and Maksim Horowitz, "nflWAR: A reproducible method for offensive player evaluation in football," *Journal of Quantitative Analysis in Sports* 15, no. 3 (2019): 163-183, <https://doi.org/10.1515/jqas-2018-0010>.

³nflfastR documentation (authors Sebastian Carl and Ben Baldwin): <https://www.nflfastR.com/>; Ben Baldwin, "nflfastR EP, WP, and CP models," Open Source Football, September 28, 2020: <https://www.opensourcefootball.com/posts/2020-09-28-nflfastR-ep-wp-and-cp-models/>; the nflverse organisation: <https://github.com/nflverse>.

⁴nflverse, nfl_data_py (archived repository; README: deprecated in favour of nlfireadpy, no further maintenance): https://github.com/nflverse/nfl_data_py.

⁵nflverse, nlfireadpy: <https://github.com/nflverse/nlfireadpy>; version 0.1.5 released on PyPI on November 19, 2025 (<https://pypi.org/project/nlfireadpy/>), per the course fact sheet, checked October 2026.

The chapter's three pure illustrations (the flow chart below, the map of a row's columns and the data timeline) are drawn by a few helper functions. Their code is about boxes and arrows, not football or data, so it is folded away in the web version and left out of the printed one. Every cell that touches data is shown.

Here is how the pieces fit together, from the stadium to the pictures in this book:

```
draw_flow()
plt.show()
```

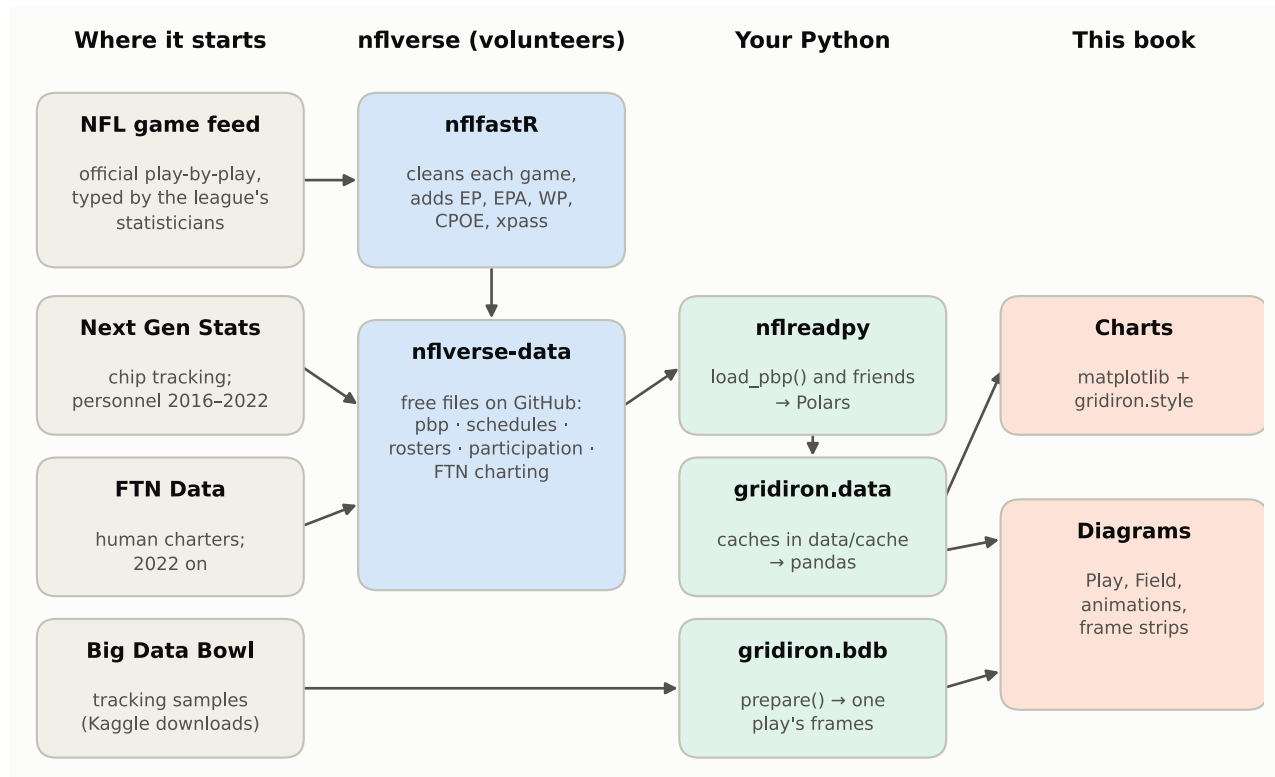


Figure 1: Where this book's numbers and pictures come from. Left: the three suppliers nflverse draws on (the league's official play-by-play, Next Gen Stats tracking, and FTN's human charters), plus the Big Data Bowl's tracking samples, which you download from Kaggle yourself. nflfastR turns the league's record into a clean table and adds the model columns; nflverse publishes everything as free files. In Python, nflreadpy downloads them as Polars data frames, and the course's gridiron.data wrapper caches them and hands over pandas. Charts use gridiron.style; diagrams and animations use gridiron's Play and Field, fed either by a hand-drawn play or by real tracking frames through gridiron.bdb.

Two things in that picture are worth holding on to. First, nflverse is a **republisher**, not a source: it does not chart games itself. The play-by-play reflects what the league's statisticians wrote; the personnel and formation data reflect what Next Gen Stats or FTN's charters recorded. When those suppliers make a mistake, or change how they record something, nflverse inherits it. Second, the **models are nflverse's own**. When this book quotes EPA or win probability, it is quoting nflfastR's models, which are good, public and documented, but are one model among several. ESPN's win probability on a broadcast is a different model and will not match to the decimal.

! Why it exists

Before free play-by-play, serious football analysis lived inside teams and a few paid services. You could read that "teams pass too much on first down" but you could not check it. A public, reproducible table changes the argument: anyone can run the numbers, find the flaw in someone else's, and publish a better version. The cost is that you inherit the data's quirks, and this chapter is mostly about those quirks. The trade is worth it.

Installing nflreadpy and loading your first season

On your own machine, the whole setup is one install and two lines of Python:

```
# In a terminal:  pip install nflreadpy pandas pyarrow      (or: uv add nflreadpy pandas pyarrow)
import nflreadpy as nfl

plays = nfl.load_pbp([2025])      # a Polars DataFrame, one row per play
plays_pd = plays.to_pandas()      # the same data as pandas (needs pyarrow)
```

The first call downloads the season from nflverse’s GitHub releases, a file of a few tens of megabytes, so it takes a few seconds. Every loader takes a list of seasons, a single season, or `True` for everything available.

This book wraps `nflreadpy` in a tiny module, `gridiron.data`, so that every chapter shares one cache on disk. `nfl()` returns the configured `nflreadpy` module and `pbp()` loads seasons and hands back pandas. The cells below use it; on your machine you can call `nflreadpy` directly.

```
nflr = nfl()                      # the nflreadpy module, pointed at data/cache
raw25 = nflr.load_pbp([2025])     # Polars
print(type(raw25).__name__, f"{raw25.height:,} rows x {raw25.width} columns")
```

DataFrame 48,771 rows x 372 columns

Polars or pandas?

`nflreadpy` returns **Polars** data frames. Polars is a newer data-frame library, written in Rust, that stores each column as one block of memory and runs operations in parallel; on a ten-season load it is noticeably faster than pandas and uses less memory. Most Python tutorials, and most of this book, use pandas, because it is what most readers know. You can work in either. Here is the same question, “how often did offenses pass on each down in the 2025 regular season?”, asked both ways:

```
# Polars: chain expressions on the frame nflreadpy returned
by_down_pl = (
    raw25.filter((pl.col("season_type") == "REG")
                 & pl.col("play_type").isin(["pass", "run"])
                 & pl.col("down").is_not_null())
    .group_by("down")
    .agg(pl.col("pass").mean().alias("pass_rate"), pl.len().alias("plays"))
    .sort("down")
)
print(by_down_pl)

# pandas: convert once, then the familiar idioms
df25 = raw25.to_pandas()
reg25 = df25[(df25.season_type == "REG") & df25.play_type.isin(["pass", "run"])
             & df25.down.notna()]
print(reg25.groupby("down")["pass"].agg(pass_rate="mean", plays="size").round(3))
```

shape: (4, 3)

down	pass_rate	plays
---	---	---
f64	f64	u32
1.0	0.497722	14267
2.0	0.617417	10863

3.0	0.785032	6801
4.0	0.6678	882

	pass_rate	plays
down		
1.0	0.498	14267
2.0	0.617	10863
3.0	0.785	6801
4.0	0.668	882

The answers are identical, as they should be. Two habits make the conversion painless. Convert **after** you have selected the columns you need (`raw25.select([...]).to_pandas()`), because a 372-column frame is mostly columns you will never touch; `gridiron.data.pbp(seasons, columns=[...])` does exactly that. And remember that Polars has no index: if pandas code does `.loc[...]` on the result of a groupby, it is relying on the index pandas created, which Polars never had.

The cache, and the current season

`nflreadpy` keeps what it downloads. By default the cache lives **in memory** for 24 hours, so a second load in the same session is instant and a new session downloads again. The course instead caches **on disk** for a year, so the book can be rebuilt without fetching ten seasons every time:

```
from nflreadpy.config import update_config

update_config(cache_mode="filesystem", cache_dir="data/cache",
              cache_duration=60 * 60 * 24 * 365)  # seconds; the course keeps files a year
```

A year is right for completed seasons, which change rarely. It is wrong for the season in progress. `nflverse` refreshes the current season's play-by-play as games are played, so during the season you want a short cache (say `cache_duration=6 * 60 * 60`, six hours) or a fresh download (`nflreadpy.clear_cache()`). The Sunday notebook at the end of the chapter does exactly that.

Common misconception

“The Python package for NFL data is `nfl_data_py`.” It was, and thousands of tutorials and notebooks still use it (`import nfl_data_py as nfl; nfl.import_pbp_data([2023])`). It is now archived and unmaintained, so it will drift away from the data as `nflverse` changes its files. Use `nflreadpy`: the function names start with `load_` instead of `import_`, it returns Polars instead of pandas, and otherwise the data is the same `nflverse` data.

Anatomy of a play-by-play row

Here is Walker's run as `nflverse` records it, a handful of its 372 columns:

```
walker = df25[(df25.game_id == "2025_22_SEA_NE") & (df25.play_id == 1015)].iloc[0]
MEANING = {"game_id": "season _ week _ away team _ home team", "play_id": "the play's number in the game",
           "qtr": "quarter", "time": "game clock at the snap", "posteam": "team with the ball",
           "down": "down", "ydstogo": "yards to the line to gain",
           "yardline_100": "yards to the opponent's goal line", "play_type": "kind of play",
           "yards_gained": "yards gained on the play", "ep": "expected points before the snap",
           "epa": "expected points added by the play",
           "wp": "offense's win probability before the snap"}
FMT = {"play_id": "{:.0f}", "qtr": "{:.0f}", "down": "{:.0f}", "ydstogo": "{:.0f}",
       "yardline_100": "{:.0f}", "yards_gained": "{:.0f}", "ep": "{:.2f}", "epa": "{:.2f}",
       "wp": "{:.2f}"}
md_table(pd.DataFrame({"column": list(MEANING),
                      "value": [FMT.get(c, "{}").format(walker[c]) for c in MEANING],
                      "meaning": list(MEANING.values())}))
```

column	value	meaning
game_id	2025_22_SEA_NE	season _ week _ away team _ home team
play_id	1015	the play's number in the game
qtr	2	quarter
time	14:04	game clock at the snap
posteam	SEA	team with the ball
down	2	down
ydstogo	10	yards to the line to gain
yardline_100	76	yards to the opponent's goal line
play_type	run	kind of play
yards_gained	30	yards gained on the play
ep	0.12	expected points before the snap
epa	+2.62	expected points added by the play
wp	0.53	offense's win probability before the snap

Read it left to right and it is a scorebug plus a result. `game_id` names the game (season, week, away team, home team: Seattle was the designated away team), and `play_id` numbers the play within it. Seattle (`posteam`, the team with the ball) has second down (`down`) and 10 to go (`ydstogo`) at its own 24, which `nflverse` writes as `yardline_100 = 76`; the next section explains why. The play was a run that gained 30 yards. Before the snap the situation was worth 0.12 expected points (`ep`); the run added 2.62 (`epa`), because it turned second-and-10 deep in Seattle's half into first-and-10 in New England's. Before the snap, `nflfastR`'s model gave Seattle a 53% chance of winning (`wp`), the model's estimate from the score, the clock, the field position and the pre-game point spread.

The other 359 columns fall into a few families. You will use perhaps forty of them regularly, so it pays to know which family to look in:

```
KEYS = {"play_id", "game_id", "old_game_id", "home_team", "away_team", "season_type", "week",
        "game_date", "season", "nfl_api_id", "order_sequence"}
SITUATION = {"posteam", "posteam_type", "defteam", "side_of_field", "yardline_100", "yrdln",
             "quarter_seconds_remaining", "half_seconds_remaining", "game_seconds_remaining",
             "game_half", "qtr", "down", "goal_to_go", "time", "ydstogo", "home_timeouts_remaining",
             "away_timeouts_remaining", "posteam_timeouts_remaining",
             "defteam_timeouts_remaining", "total_home_score", "total_away_score",
             "posteam_score", "defteam_score", "score_differential"}
MODEL = r"^(^|_)(ep|epa|wp|wpa|cp|cpoe|ypass|pass_oe)$|_prob$|^xyac|_epa$|_wpa$|^vegas|_wp$"
PLAYERS = r"player_id|player_name|_player_|jersey|^(passer|rusher|receiver|name|id)$|_id$|fantasy"
GAME = (r"^(drive|^fixed_drive|^series|stadium|roof|surface|^temp$|^wind$|weather|_line$|coach"
        r"|^(location|result|total|div_game|start_time|time_of_day|home_score|away_score)$"
        r"|home_opening_kickoff|end_clock_time|end_yard_line")

def family(col):
    if col in KEYS:
        return "Keys"
    if col in SITUATION:
        return "Situation before the snap"
    if re.search(MODEL, col):
        return "Model estimates"
    if re.search(GAME, col):
        return "Drive and game"
```

```

if re.search(PLAYERS, col):
    return "Players"
return "What happened"

fam = pd.Series({c: family(c) for c in raw25.columns})
counts = fam.value_counts()
print(counts.to_string())

w = walker
EXAMPLES = {
    "Keys": [("game_id", w.game_id), ("play_id", f"{w.play_id:.0f}"), ("season", w.season),
            ("week", w.week), ("season_type", w.season_type)],
    "Situation before the snap": [("posteam", w.posteam),
                                ("qtr · time", f"{w.qtr:.0f} · {w.time}"),
                                ("down", f"{w.down:.0f}"), ("ydstogo", f"{w.ydstogo:.0f}"),
                                ("yardline_100", f"{w.yardline_100:.0f}"),
    "What happened": [("play_type", w.play_type), ("yards_gained", f"{w.yards_gained:.0f}"),
                    ("shotgun", f"{w.shotgun:.0f}"), ("run_location", w.run_location),
                    ("first_down", f"{w.first_down:.0f}"),
    "Model estimates": [("ep", f"{w.ep:.2f}"), ("epa", f"{w.epa:+.2f}"), ("wp", f"{w.wp:.2f}"),
                      ("wpa", f"{w.wpa:+.3f}"), ("xpass", f"{w.xpass:.2f}"),
    "Players": [("rusher", w.rusher), ("rusher_jersey_number", f"{w.rusher_jersey_number:.0f}"),
              ("rusher_player_id", w.rusher_player_id), ("passer", "(none)",
              ("...", "≈100 more")],
    "Drive and game": [("drive", f"{w.drive:.0f}"), ("series", f"{w.series:.0f}"),
                    ("fixed_drive_result", w.fixed_drive_result),
                    ("spread_line", f"{w.spread_line:+.1f}"), ("roof", w.roof)],
}
draw_schema(EXAMPLES, counts, w.desc)
plt.show()

```

What happened	125
Players	102
Model estimates	64
Drive and game	46
Situation before the snap	24
Keys	11

Keys 11 columns game_id 2025_22_SEA_NE play_id 1015 season 2025 week 22 season_type POST	Situation before the snap 24 columns posteam SEA qtr · time 2 · 14:04 down 2 ydstogo 10 yardline_100 76	What happened 125 columns play_type run yards_gained 30 shotgun 1 run_location left first_down 1
Model estimates 64 columns ep 0.12 epa +2.62 wp 0.53 wpa +0.077 xpass 0.61	Players 102 columns rusher K.Walker rusher_jersey_number 9 rusher_player_id 00-0038134 passer (none) ... ≈100 more	Drive and game 46 columns drive 7 series 14 fixed_drive_result Field goal spread_line -4.5 roof outdoors
desc (14:04) (Shotgun) 9-K.Walker left end pushed ob at NE 46 for 30 yards (25-M.Jones).		

Figure 2: The 372 columns of a 2025 play-by-play row, sorted into six families (counts from the code below; the sorting is the book's, not nflverse's), with Walker's 30-yard run from Super Bowl LX as the example row. Keys identify the play; the situation is what the scorebug showed before the snap; what happened is the result; the model columns are nflfastR's estimates; the player columns name and identify everyone involved; the drive and game columns describe the possession and the game around it. The play description at the bottom is the original text the league's statisticians wrote.

Two things about the families matter in practice. The **situation** columns are the scorebug, so they are what you filter on. The **model** columns are where the analysis is: `ep` and `epa` for value, `wp` and `wpa` for winning, `cp` and `cpoe` for passing accuracy, `xpass` and `pass_oe` for play-calling tendencies. They are estimates, not facts; Expected Points, EPA, and Success Rate builds the first of those models from scratch, and Win Probability and Decision Analysis the second.

i Madden vs. real life

In Madden every play has a name: you pick "Inside Zone" or "PA Boot Over" from a menu, and the game knows what you called. Real play-by-play never records the call. It records what happened ("left end for 30 yards"), in the official statisticians' words, and nothing about whether it was outside zone, a toss or a run-pass option. Even the flags that sound like scheme (`shotgun`, `no_huddle`, `run_location`) are describing the result, not reading the playbook. The scheme labels you'll meet later in this chapter come from human charters watching film, and they are judgement calls.

A row is not always a play

The table has one row per *event* in the league's log, and most events are not snaps. Here is the 2025 file by `play_type`:

```
pt = (df25.play_type.fillna("(none)").value_counts().rename_axis("play_type")
      .reset_index(name="rows"))
pt["share"] = pt.rows / pt.rows.sum()
md_table(pt, digits=3)
```

play_type	rows	share
pass	19,737	0.405
run	14,895	0.305
no_play	4,723	0.097
kickoff	2,927	0.060
punt	2,042	0.042
(none)	1,445	0.030
extra_point	1,330	0.027
field_goal	1,140	0.023
qb_kneel	452	0.009
qb_spike	80	0.002

Only about 70% of the rows are runs and passes. The rest are kickoffs, punts, extra points and field goals; `no_play`, which is either a play wiped out by an accepted penalty or a timeout; kneel- downs and spikes; and rows with no `play_type` at all, which are bookkeeping: the start of the game, the end of each quarter, the two-minute warning. Every analysis starts by deciding which of these to keep.

yardline_100: one number for field position

`yardline_100` is the number of yards from the ball to the goal line the offense is attacking. Your own 25 is 75; midfield is 50; the opponent's 10 is 10. The broadcast's way of saying it, "SEA 24," needs you to know which team has the ball and which way it is going; `yardln`, the column that stores the broadcast form, has the same problem. `yardline_100` doesn't:

```
start, gain = float(walker.yardline_100), float(walker.yards_gained)
x0, x1 = 110 - start, 110 - (start - gain)          # BDB x for an offense moving +x
fld = Field(orient="horizontal", los=x0, window=(-x0, 120 - x0), to_go=float(walker.ydstogo),
            width_in=6.5)
ax = fld.ax
yb = 22.0                                           # ball on the near side of the field
ax.add_patch(FancyArrowPatch((x0, yb), (x1, yb), arrowstyle="->|>", mutation_scale=12,
                             color=style.OFFENSE_DARK, lw=2.4, zorder=5))
ax.add_patch(Circle((x0, yb), 1.0, color=style.BALL, zorder=6))
BOX = dict(boxstyle="round,pad=0.3", fc=style.SURFACE, ec=style.BASELINE, lw=0.6)
ax.text(x0, 33.0, f"yardln: 'SEA {110 - start - 10:.0f}'\nyardline_100: {start:.0f}\nBDB x: {x0:.0f}",
        ha="center", va="center", fontsize=7, color=style.INK, bbox=BOX, zorder=8)
ax.text(x1, 33.0, f"yardln: 'NE {start - gain:.0f}'\nyardline_100: {start - gain:.0f}\n"
        f"BDB x: {x1:.0f}", ha="center", va="center", fontsize=7, color=style.INK, bbox=BOX,
        zorder=8)
ax.text((x0 + x1) / 2, yb + 1.6, f"+{gain:.0f} yards", ha="center", va="bottom", fontsize=7.5,
        fontweight="bold", color=style.OFFENSE_DARK, zorder=8)
for gx, txt in [(5, "Seattle's\nend zone"), (115, "New England's\nend zone")]:
    ax.text(gx, 26.7, txt, ha="center", va="center", fontsize=6.5, color=style.INK_2,
            rotation=90, zorder=8)
for yl in range(100, -1, -10):                    # yardline_100 ruler near the far sideline
    ax.text(110 - yl, 3.2, f"{yl}", ha="center", va="center", fontsize=6.3,
            color=style.OFFENSE_DARK, zorder=8)
ax.text(60, 6.6, "yardline_100", ha="center", va="center", fontsize=6.5, style="italic",
        color=style.OFFENSE_DARK, zorder=8)
ax.text(52, 47.5, "Seattle moving this way →", ha="center", va="center", fontsize=7.5,
        color=style.INK_2, zorder=8)
plt.show()
```

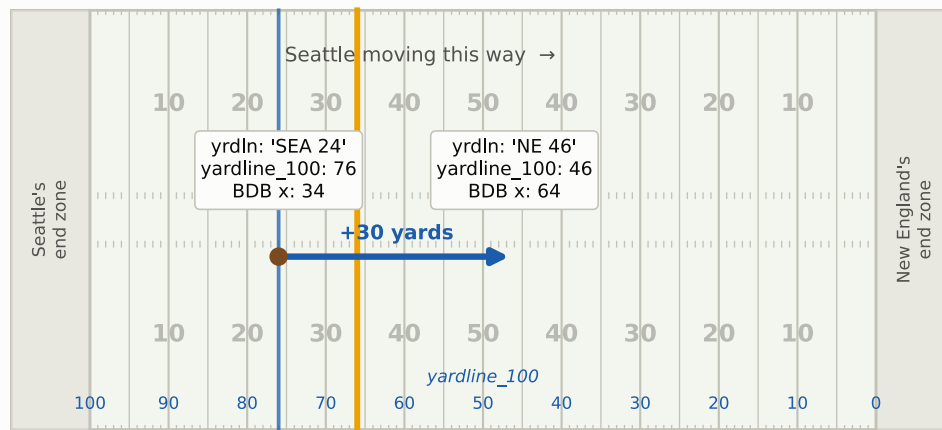


Figure 3: Walker’s 30-yard run on the field, with Seattle moving left to right. The broadcast and the yrdln column call the starting spot ‘SEA 24’; yardline_100 calls it 76, the yards still to go for a touchdown; the Big Data Bowl tracking frame (offense normalized to move toward higher x) puts it at $x = 34$, since $x = 110 - \text{yardline_100}$. The run ends at ‘NE 46’: $\text{yardline_100} = 46$, $x = 64$. The yellow line is the line to gain. The bottom ruler shows yardline_100 from 100 at Seattle’s goal line to 0 at New England’s.

! Why it exists

Football’s spoken language for field position is relative (“own 24,” “plus-46,” “the 30 going in”), and the same words mean different spots for the two teams. Every computation that touches field position (expected points, red-zone filters, “how far from the end zone did this drive start?”) needs a single scale that runs in the offense’s direction. `yardline_100` is that scale. `yardline_100 <= 20` is the red zone for whichever team has the ball, and `110 - yardline_100` puts the ball on the Big Data Bowl’s x axis for an offense moving toward higher x , which is how this book’s diagrams are drawn. The trade-off is that it is useless for “which end of the stadium?” questions (wind, sun, the home crowd), for which you need `yrdln` and `posteam` together.

Flags, not categories: `pass`, `rush` and `play_type`

`play_type` sounds like the answer to “was it a run or a pass?” It isn’t quite. nflverse files a **scramble** (a quarterback who drops back to throw, finds no one open and runs) as `play_type == "run"`, because the ball was carried, not thrown. A **sack** is `play_type == "pass"`. The 0/1 flag columns tell you what the offense *intended*:

```
flagged = df25[df25.season_type == "REG"].copy()
flagged["kind"] = np.select(
    [flagged.qb_scramble == 1, flagged.sack == 1, flagged.qb_kneel == 1, flagged.qb_spike == 1,
     flagged.two_point_attempt == 1, flagged.play_type == "no_play",
     flagged.play_type == "pass", flagged.play_type == "run"],
    ["scramble", "sack", "kneel-down", "spike", "two-point try", "no-play",
     "pass", "designed run"], "other")
keep = ["designed run", "pass", "sack", "scramble", "two-point try", "kneel-down", "spike",
        "no-play"]
flag_tab = (flagged[flagged.kind.isin(keep)]
            .groupby("kind")
            .agg(plays=("play_id", "size"), play_type=("play_type", lambda s: s.mode()[0]),
                 has_down=("down", lambda s: s.notna().mean()), pass_flag=("pass", "mean"),
                 rush_flag=("rush", "mean"), qb_dropback=("qb_dropback", "mean"))
            .reindex(keep).reset_index())
md_table(flag_tab)
```

kind	plays	play_type	has_down	pass_flag	rush_flag	qb_dropback
designed run	13,079	run	1.00	0.00	1.00	0.00

kind	plays	play_type	has_down	pass_flag	rush_flag	qb_dropback
pass	17,358	pass	1.00	1.00	0.00	1.00
sack	1,287	pass	1.00	1.00	0.00	1.00
scramble	1,160	run	1.00	1.00	0.00	0.94
two-point try	128	pass	0.00	0.76	0.24	0.75
kneel-down	434	qb_kneel	1.00	0.00	0.00	0.00
spike	81	qb_spike	1.00	0.00	0.00	0.00
no-play	4,464	no_play	0.54	0.24	0.07	0.00

Read the table row by row and the logic is clear. `pass == 1` marks every **dropback**: real passes, sacks and scrambles, because each began as a pass play. `rush == 1` marks only **designed runs**, including quarterback runs that were called as runs. Two-point tries have **no down**, so a `down.notna()` filter quietly drops them, which is usually what you want. Kneel-downs and spikes get their own `play_type` and neither flag. And `no_play` rows sometimes carry a flag (the run that was called back for holding is still marked as a run) and sometimes don't (a timeout).

This is why this book counts a "pass" as `pass == 1`, and says so in its captions: "sacks and scrambles count as passes." It is the honest measure of what the offense called. Count passes with `play_type == "pass"` and you lose the scrambles, about a thousand plays a season, and every pass rate comes out a couple of points low.

Filtering to "real" plays

Almost every question in this book starts from the same funnel: the regular season, runs and dropbacks only, with a down, and (for questions about play-calling and efficiency) without garbage time. Here is what each step removes from the 2025 file:

```
s0 = df25
s1 = s0[s0.season_type == "REG"]
s2 = s1[s1.play_type.isin(["pass", "run"])]
s3 = s2[s2.down.notna()]
s4 = s3[s3.wp.between(0.10, 0.90)]
s5 = s4[s4.half_seconds_remaining > 120]
STEPS = [("Every row in the 2025 file", s0, ""),
         ("Regular season only", s1, "playoff games"),
         ("Runs and passes (play_type)", s2, "kicks, no-plays, kneels, spikes, timeouts"),
         ("...that have a down", s3, "two-point tries"),
         ("Win probability 10-90%", s4, "lopsided games"),
         ("Not in the last 2 min of a half", s5, "two-minute drills")]
fig, ax = plt.subplots(figsize=(6.5, 2.7))
ax.set_axisbelow(True)
ys = np.arange(len(STEPS))[:-1]
for y, (lab, frame, why) in zip(ys, STEPS):
    gt = lab.startswith(("Win", "Not"))
    ax.barh(y, len(frame), color=style.DEFENSE if gt else style.OFFENSE, height=0.62)
    ax.text(len(frame) - 600, y, f"{len(frame):,}", ha="right", va="center", fontsize=7.5,
           color="white", fontweight="bold")
    if why:
        ax.text(len(frame) + 700, y, f"- {why}", ha="left", va="center", fontsize=7,
               color=style.INK_2)
ax.set_yticks(ys, [s[0] for s in STEPS], fontsize=7.8)
ax.set_xlim(0, 80000)
ax.set_xticks([0, 10000, 20000, 30000, 40000, 50000], ["0", "10k", "20k", "30k", "40k", "50k"],
              fontsize=7.5)
ax.set_xlabel("rows remaining", fontsize=8)
ax.grid(axis="y", visible=False)
```

```
ax.spines["left"].set_visible(False)
plt.show()
```

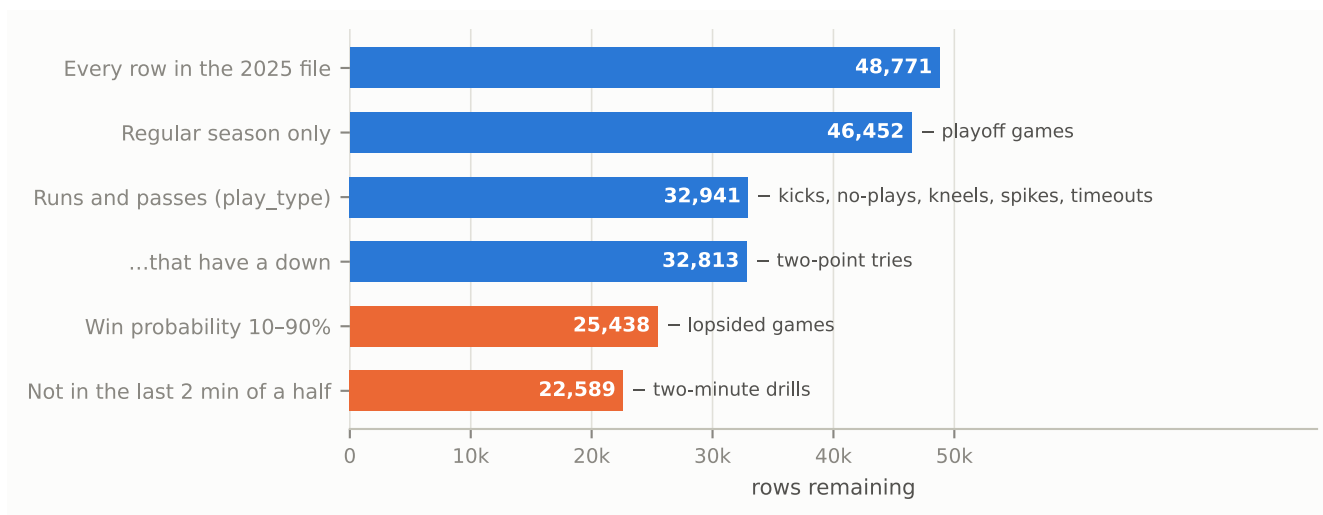


Figure 4: From every row of the 2025 file to the plays an analysis uses (nflverse play-by-play). Each bar is what survives the filter on its left; the text on the right says what that step removed. The last two steps are the garbage-time filter described below. Your own funnel may differ; the point is to know, and to say, which steps you took.

The first four steps are housekeeping. The last two are a choice, and they need explaining.

Why garbage time gets thrown out

Think about what a team does when it trails by 20 with eight minutes left. It throws on nearly every down, because runs burn clock it doesn't have, and the defense, which knows it, drops eight players into coverage and gives up underneath completions all day. Meanwhile the team that leads by 20 runs into a stacked box to keep the clock moving. Neither side is running its game plan. If you leave those plays in, a team that spent the second half of a blowout throwing will look pass-happy, and a team that spent it running out the clock will look like a running team. The score made those calls, not the coaches' preferences. Here is how strongly the score pulls on the most ordinary situation in football, first-and-10:

```
HCOLS = ["season", "season_type", "week", "game_id", "play_id", "qtr", "time", "posteam",
         "down", "ydstogo", "yardline_100", "play_type", "pass", "rush", "shotgun", "wp",
         "half_seconds_remaining", "epa", "first_down", "yards_gained", "penalty_type"]
hist = pbp(range(1999, 2026), columns=HCOLS) # 27 seasons, about 1.3 million rows
hist = hist[hist.season_type == "REG"]
snaps = hist[hist.play_type.isin(["pass", "run"]) & hist.down.notna()].copy()
```

```
f10 = snaps[snaps.season.between(2016, 2025) & (snaps.down == 1) & (snaps.ydstogo == 10)
          & (snaps.half_seconds_remaining > 120) & snaps.wp.notna()].copy()
f10["bin"] = pd.cut(f10.wp, np.linspace(0, 1, 11))
wp_tab = f10.groupby("bin", observed=True)["pass"].agg(["mean", "size"])
mids = np.arange(5, 100, 10)
fig, ax = plt.subplots(figsize=(6.5, 2.8))
ax.set_axisbelow(True)
ax.axvspan(10, 90, color=style.OFFENSE_TINT, alpha=0.6, lw=0, zorder=0)
ax.text(50, 83, "kept by the garbage-time filter", ha="center", fontsize=7.5, color=style.INK_2)
cols = [style.DEFENSE if m < 10 or m > 90 else style.OFFENSE for m in mids]
ax.bar(mids, wp_tab["mean"] * 100, width=7.6, color=cols, zorder=2)
for m, v in zip(mids, wp_tab["mean"]):
    ax.text(m, v * 100 + 1.5, f"{v:.0%}", ha="center", va="bottom", fontsize=7.3, color=style.INK)
ax.set_xticks(range(0, 101, 10), [f"{v}%" for v in range(0, 101, 10)], fontsize=7.5)
ax.set_xlim(0, 100)
```

```
ax.set_ylim(0, 90)
ax.set_xlabel("offense's win probability before the snap", fontsize=8)
ax.set_ylabel("first-and-10 pass rate (%)", fontsize=8)
ax.grid(axis="x", visible=False)
plt.show()
```

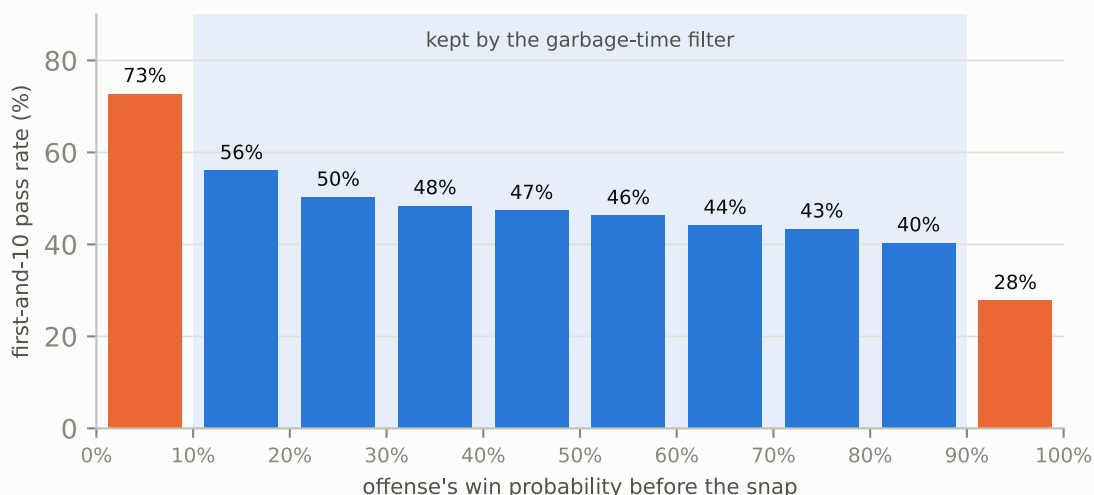


Figure 5: How often offenses passed on first-and-10, by the offense's win probability before the snap (regular seasons 2016–2025, outside the last two minutes of each half; a pass is any dropback, including sacks and scrambles). Between about 20% and 80% the pass rate barely moves. Outside that band the scoreboard takes over: teams that are almost beaten throw on about 70% of first downs, and teams that have the game won throw on fewer than three in ten. The shaded band is what this course's garbage-time filter keeps.

The middle of the chart is flat, and the ends are not. That flat middle is what the **garbage-time filter** tries to keep: plays where the game is still in doubt, so the call reflects the plan. This book's version keeps plays where the offense's win probability is **between 10% and 90%** and that are **not in the last two minutes of either half**. The win-probability part removes blowouts; the clock part removes the two-minute drill, where even a close game is played by the clock, with hurry-up passing and timeouts. Expected Points, EPA, and Success Rate uses exactly this filter for every team number.

Why 10% and 90%, and not 20% and 80%? Because there is no right answer, only a trade between keeping the plays that tell you about the game plan and keeping enough plays to measure anything. Some earlier chapters in this book (the pass-rate charts in Downs, Distance, and Field Position and Personnel) use a stricter cut, 20% to 80%, because they are about play-calling, where the score's pull starts earlier. Many public analysts, including the nflfastR guides, use similar ranges.⁶ Tendencies: PROE, Personnel, Motion, and Coverage formalises the idea as the **neutral situation**. What matters is that the choice changes the numbers, so you state it:

```
s25 = snaps[snaps.season == 2025]
late = s25.half_seconds_remaining <= 120
CHOICES = {"none": s25,
           "WP 10–90%, not last 2 min (this course)": s25[s25.wp.between(0.10, 0.90) & ~late],
           "WP 20–80%, not last 2 min (01-02, 02-01)": s25[s25.wp.between(0.20, 0.80) & ~late]}
choice_tab = pd.DataFrame({name: d.groupby("down")["pass"].mean() for name, d in CHOICES.items()}).T
choice_tab.columns = [f"down {int(c)}" for c in choice_tab.columns]
shown = choice_tab.map(lambda v: f"{v:.1%}")
shown.insert(0, "plays", [f"{len(d):,}" for d in CHOICES.values()])
md_table(shown.reset_index(names="filter (2025)"))
```

⁶nflfastR, "A beginner's guide to nflfastR" (its examples filter on win probability and on `half_seconds_remaining > 120` before computing pass rates and EPA): https://nflfastR.com/articles/beginners_guide.html.

filter (2025)	plays	down 1	down 2	down 3	down 4
none	32,813	49.8%	61.7%	78.5%	66.8%
WP 10–90%, not last 2 min (this course)	22,589	46.4%	59.6%	79.5%	56.8%
WP 20–80%, not last 2 min (01-02, 02-01)	18,426	46.0%	59.4%	79.7%	54.8%

Filtering garbage time removes about three plays in ten and moves first-down pass rate by about three points, which is the size of the differences many arguments are about. Fourth down moves most, from 67% passes to 57%, because so many fourth-down plays are desperate throws by teams that are losing. The two filters agree closely with each other; the unfiltered numbers are the odd ones out.

Common misconception

“Filtering is cheating: you’re throwing away data you don’t like.” Filtering is choosing the question. “How often does this team pass when the game is in doubt?” and “how often does this team pass?” are different questions, and both are legitimate. The cheating version is choosing the filter *after* seeing which one gives the answer you wanted, or not saying which you used. A filter stated in a caption is a method; one hidden in the code is a thumb on the scale.

Beyond play-by-play: schedules, rosters, participation and FTN

Play-by-play tells you what happened. It doesn’t tell you who was on the field, how they lined up, or whether the quarterback faked a handoff first. For that, nflverse publishes other tables, each with its own coverage in time. Knowing which seasons each one covers is half the battle:

```
draw_timeline()
plt.show()
```

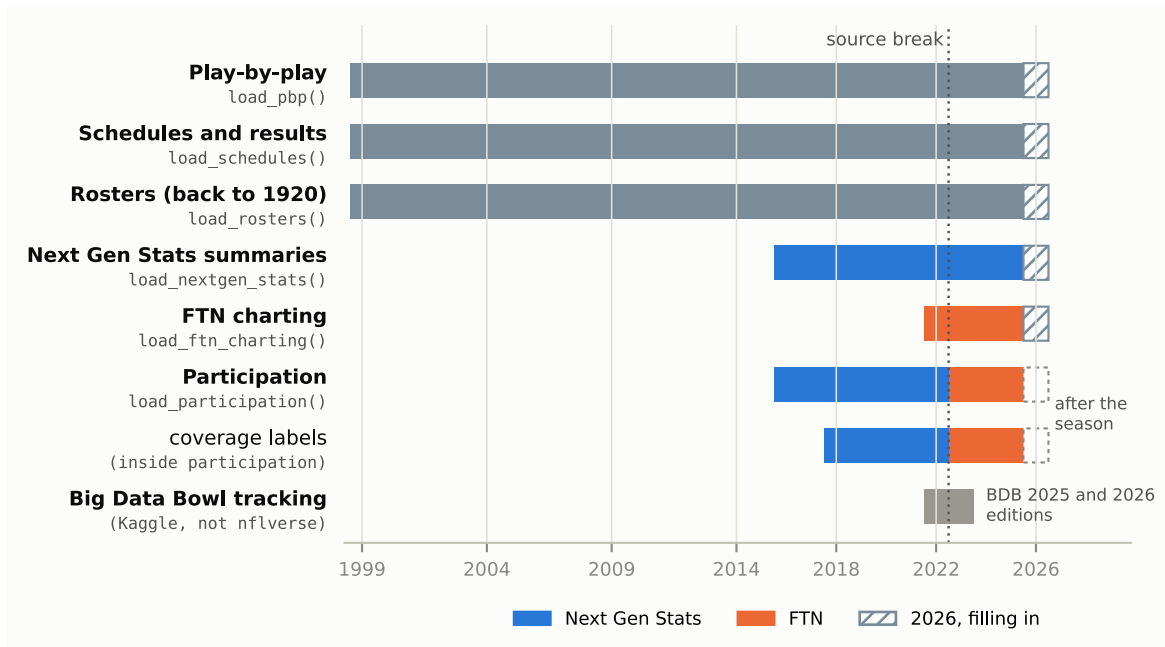


Figure 6: Which seasons each nflverse table covers, as of October 2026. Play-by-play and schedules start in 1999 (rosters go back to 1920); Next Gen Stats player summaries and participation start in 2016; FTN charting starts in 2022. Participation is supplied by Next Gen Stats (blue) through 2022 and by FTN (orange) from 2023: that is the source break. Coverage labels sit inside participation, from 2018 on. Hatched 2026 cells are being filled as the season is played; participation for 2026 arrives only after the season, around February 2027. Big Data Bowl tracking is not part of nflverse: recent editions released samples of single seasons.

Each table joins to play-by-play on a key, and the key is the first thing to learn about it.

Schedules (`load_schedules()`) have one row per game, keyed by `game_id`, with the final score, the betting line, the stadium, the roof, the weather, the starting quarterbacks, the head coaches and even the referee. They are how you attach game-level facts (was it a divisional game? on short rest?) to plays.

Rosters (`load_rosters()`) have one row per player per team per season, keyed by `gsis_id`, the league's player ID (the 00-00... strings). Play-by-play stores the same ID in `passer_id`, `rusher_id`, `receiver_id` and the other player columns, so a merge gives you a player's position, size, college or draft slot.

```

sched = nflr.load_schedules([2025]).to_pandas()
sb = sched[sched.game_id == "2025_22_SEA_NE"]
display(md_table(sb[["game_id", "gameday", "away_team", "away_score", "home_team",
                    "home_score", "spread_line", "roof"]]))

roster = nflr.load_rosters([2025]).to_pandas()
carriers = (df25[(df25.game_id == "2025_22_SEA_NE") & (df25.play_type == "run") & (df25.rush == 1)]
            .merge(roster[["gsis_id", "position", "college"]].drop_duplicates("gsis_id"),
                  left_on="rusher_player_id", right_on="gsis_id", how="left")
            .groupby(["posteam", "rusher", "position", "college"])
            .agg(carries=("play_id", "size"), yards=("yards_gained", "sum"),
                 epa=("epa", "sum"))
            .reset_index().sort_values("carries", ascending=False).head(5))
display(md_table(carriers))

```

game_id	gameday	away_team	away_score	home_team	home_score	spread_line	roof
2025_22_SEA_NE	2026-02-08	SEA	29	NE	13	-4.50	outdoors

posteam	rusher	position	college	carries	yards	epa
SEA	K.Walker	RB	Michigan State; Wake Forest	27	135	-2.04
NE	R.Stevenson	RB	Oklahoma; Cerritos College	7	23	-1.60
NE	T.Henderson	RB	Ohio State	6	19	-1.31
SEA	G.Holani	RB	Boise State	2	6	0.17
SEA	R.Shaheed	WR	Weber State	1	-5	-1.28

(Seattle was the designated road team, and `spread_line` is in the home team's favour, so -4.5 means the "home" Patriots were 4.5-point underdogs.) Walker's line from the merge matches the box score (27 carries, 135 yards), which is the first check you should always make after a join: does a total you already know come out right? It also holds a small surprise. The game's MVP totals slightly *negative* EPA on his carries, because a run of short gains on early downs each cost a little, and the 30- and 29-yard bursts did not quite pay for them. That is the gap between yards and value that Expected Points, EPA, and Success Rate is about.

Participation data (`load_participation()`) is the table that powers the personnel and formation charts in Part 2. It has one row per play, keyed by `nflverse_game_id` and `play_id`, and lists the offensive and defensive personnel, the formation, the number of defenders in the box, the number of pass rushers, the players' IDs and, from 2023, their names, positions and jersey numbers. For most dropbacks from 2018 on it also carries the coverage: `defense_man_zone_type` (man or zone) and `defense_coverage_type` (COVER_1, COVER_3 and so on). Two facts about it shape everything you do with it. It comes from two suppliers, Next Gen Stats for 2016–2022 and FTN from 2023. And it is published **only after a season ends**: FTN releases each season's file under a Creative Commons licence (CC-BY-SA 4.0) once the season is over, so the 2026 file will not exist until about February 2027.⁷

FTN charting data (`load_ftn_charting()`) is the other half of FTN's work: a human charter's notes on every play, from 2022 on, updated **weekly during the season**. It records where the quarterback lined up (`qb_location`: U for under center, S for shotgun, P for pistol), how many players were in the backfield and in the box, whether there was motion, play-action, a screen or an RPO, how many defenders blitzed, and judgement calls on the throw (catchable? contested? dropped? interception-worthy?). It joins on `nflverse_game_id` and `nflverse_play_id`. It does not have the coverage; that lives only in participation.⁸

Two definitions in it trip people up. `is_motion` means motion **before or at** the snap, not only motion at the snap, so it is not the same thing as the "motion at the snap" rates you'll hear on broadcasts, nor the same as PFF's shift-and-motion rate. And there is no flag for the tush push: use `is_qb_sneak`, which covers both the classic sneak and the pushed version.

Joining participation, and the 2022-to-2023 source break

Joining is one merge. The care is in the checking. Here are ten seasons of runs and dropbacks joined to participation, with the match rate by season:

```
SEASONS = list(range(2016, 2026))
part = nflr.load_participation(SEASONS).to_pandas()
plays = snaps[snaps.season.isin(SEASONS)]
joined = plays.merge(part, left_on=["game_id", "play_id"],
                    right_on=["nflverse_game_id", "play_id"], how="left", indicator=True)
check = joined.groupby("season").agg(
    plays=("play_id", "size"),
    matched=("merge", lambda s: (s == "both").mean()),
    personnel=("offense_personnel", lambda s: s.notna().mean()),
    formation=("offense_formation", lambda s: s.notna().mean()),
    coverage=("defense_coverage_type", lambda s: s.notna().mean()))
md_table(check.reset_index(), digits=3)
```

⁷nflverse, participation data dictionary: https://nflreadr.nflverse.com/articles/dictionary_participation.html; release notes for the FTN-sourced files (2023 and 2024 re-released in September 2025, 2025 in February 2026) at <https://github.com/nflverse/nflverse-data/releases>; per the course fact sheet, checked October 2026.

⁸nflverse, FTN charting data dictionary: https://nflreadr.nflverse.com/articles/dictionary_ftn_charting.html (`is_motion`: "motion occurred on the play before or at the time of the snap"). FTN Data: <https://ftndata.com/>.

season	plays	matched	personnel	formation	coverage
2016	32,291	1.000	1.000	0.990	0.000
2017	31,987	1.000	1.000	0.992	0.000
2018	31,751	1.000	1.000	0.991	0.551
2019	32,047	1.000	1.000	0.999	0.553
2020	32,444	1.000	1.000	0.996	0.551
2021	33,986	1.000	1.000	0.993	0.547
2022	33,652	1.000	1.000	0.999	0.534
2023	33,836	1.000	1.000	0.997	0.610
2024	33,335	1.000	1.000	0.999	0.603
2025	32,813	1.000	1.000	0.998	0.601

Every run and dropback finds its participation row, and almost all of them have personnel and a formation. Coverage is missing for 2016–2017 entirely and present on a little over half of runs and dropbacks after that, because it is charted for dropbacks only. So far, so good. Now look at *what* the formation column says, season by season:

```
form = joined[joined.offense_formation.notna()].copy()
form["label"] = form.offense_formation.replace({"JUMBO": "JUMBO / WILDCAT",
                                                "WILDCAT": "JUMBO / WILDCAT"})
ORDER = ["SHOTGUN", "EMPTY", "PISTOL", "SINGLEBACK", "I_FORM", "JUMBO / WILDCAT", "UNDER CENTER"]
FCOL = {"SHOTGUN": style.OFFENSE, "EMPTY": "#6da7ec", "PISTOL": style.THIRD,
        "SINGLEBACK": "#eda100", "I_FORM": "#e87ba4", "JUMBO / WILDCAT": "#4a3aa7",
        "UNDER CENTER": style.DEFENSE}
fshare = pd.crosstab(form.season, form.label, normalize="index").reindex(columns=ORDER,
                                                                    fill_value=0) * 100

fig, ax = plt.subplots(figsize=(6.5, 3.2))
ax.set_axisbelow(True)
bottom = np.zeros(len(fshare))
for lab in ORDER:
    v = fshare[lab].values
    ax.bar(fshare.index, v, bottom=bottom, color=FCOL[lab], width=0.72, label=lab,
          edgcolor=style.SURFACE, lw=0.5)
    for x, b, h in zip(fshare.index, bottom, v):
        if h >= 6:
            ax.text(x, b + h / 2, f"{h:.0f}", ha="center", va="center", fontsize=6.5,
                  color="white" if lab not in ("SINGLEBACK", "EMPTY") else style.INK)
    bottom += v
ax.axvline(2022.5, color=style.INK, lw=1, ls=(0, (1, 2)))
ax.set_xticks(fshare.index, [str(s) for s in fshare.index], fontsize=7.5)
ax.set_ylim(0, 100)
ax.set_ylabel("share of runs and dropbacks (%)", fontsize=8)
ax.grid(axis="x", visible=False)
ax.legend(fontsize=6.6, ncol=4, loc="upper center", bbox_to_anchor=(0.5, -0.1), frameon=False)
plt.show()
```

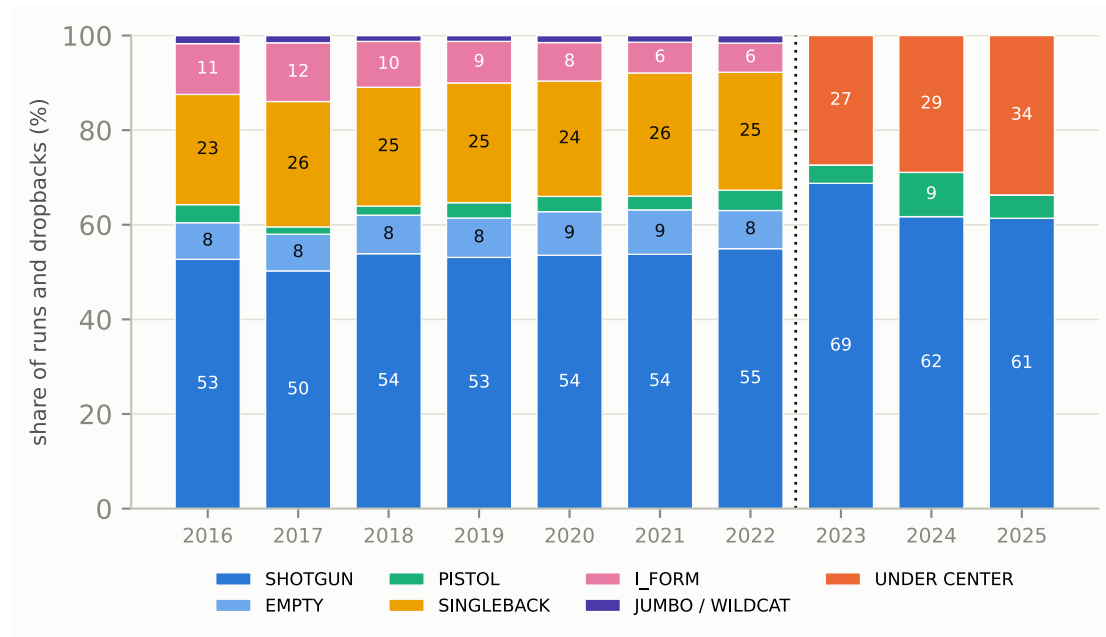


Figure 7: The offense_formation labels in nflverse participation data, share of regular-season runs and dropbacks by season. Through 2022 (Next Gen Stats) the labels describe the backfield: shotgun, singleback, I-formation, empty, pistol, jumbo, wildcat. From 2023 (FTN) there are only three, and they describe where the quarterback stood: shotgun, under center, pistol. Empty backfields were folded into shotgun, which is why ‘shotgun’ jumps from 55% to 69% across the dotted line. The football did not change that much in one off-season; the labels did.

That chart is a **source break**: a point in a series where the supplier or the method of recording changes, so the two sides are not directly comparable. Nothing in the football explains a formation vocabulary shrinking from seven words to three over one off-season. Next Gen Stats described the backfield; FTN describes the quarterback’s spot. An I-formation and a singleback set are both “UNDER CENTER” now, and an empty backfield, in which the quarterback is nearly always in the shotgun, is now “SHOTGUN.”

The break shows up in other columns too, and each one has its own trap:⁹

- **Personnel strings change format.** Through 2022 they read "1 RB, 2 TE, 2 WR", listing linemen only when there are more than five. From 2023 they list every player by roster position, "1 C, 2 G, 1 QB, 1 RB, 2 T, 2 TE, 2 WR", so any code that splits the string has to handle both, as the reproduction below does.
- **Coverage labels change.** FTN added COVER_9, COMBO and BLOWN, and dropped PREVENT.
- **The man-zone split jumps** in a way that looks like coding, not football:

```
mz = joined[(joined["pass"] == 1) & joined.defense_man_zone_type.isin(["MAN_COVERAGE",
                                                                        "ZONE_COVERAGE"])]
man = mz.groupby("season").defense_man_zone_type.apply(lambda s: (s == "MAN_COVERAGE").mean())
print((man * 100).round(1).to_string())
```

```
season
2018    35.0
2019    37.2
2020    34.4
2021    30.3
2022    28.8
2023    41.2
2024    48.3
2025    30.8
```

⁹Course calculation from nflverse participation data, regular-season runs and dropbacks 2016-2025 (formation labels, personnel strings, coverage labels and man share by season), matching the course fact sheet’s checks of October 2026.

Man coverage on 29% of labelled dropbacks in 2022, then 41% and 48% under the new supplier, then back to 31% in 2025. Defenses did not double their man coverage and then abandon it; the charters changed how they called borderline coverages. A chart that ran straight across 2022–2025 would invent a trend. (Tendencies: PROE, Personnel, Motion, and Coverage builds coverage charts with this caveat built in.)

Three rules for a source break

1. **Mark it.** Any chart that crosses 2022–2023 in participation data gets a dotted line and a note, as the personnel chart below does. A reader can then judge the jump for themselves.
2. **Translate to what both sides can say.** Before and after the break, both suppliers could count running backs and tight ends, so personnel survives the break; the formation labels don't. When the labels disagree, drop to the coarser category both can express, or find another column that does not break.
3. **Cross-check with an independent source.** Play-by-play has its own `shotgun` flag, parsed from the statisticians' play description, and FTN's `qb_location` is a separate charter's call. If they agree with each other across the years, the trend is real.

Here is rule 3 at work on the “under-center comeback” that this book describes in The Modern Era:

```
ftn = nflr.load_ftn_charting([2022, 2023, 2024, 2025]).to_pandas()
ftn["qb_location"] = ftn.qb_location.astype(str).str.strip()
fj = plays[plays.season >= 2022].merge(
    ftn[["nflverse_game_id", "nflverse_play_id", "qb_location", "is_motion", "is_rpo"]],
    left_on=["game_id", "play_id"], right_on=["nflverse_game_id", "nflverse_play_id"],
    how="left")
xc = pd.DataFrame({
    "pbp: shotgun flag": plays.groupby("season").shotgun.mean(),
    "FTN: under center (U)": fj.groupby("season").qb_location.apply(lambda s: (s == "U").mean()),
    "FTN: pistol (P)": fj.groupby("season").qb_location.apply(lambda s: (s == "P").mean())})
md_table(xc.map(lambda v: f"{v:.1%}" if pd.notna(v) else "-").reset_index())
```

season	pbp: shotgun flag	FTN: under center (U)	FTN: pistol (P)
2016	64.2%	–	–
2017	59.3%	–	–
2018	63.9%	–	–
2019	64.7%	–	–
2020	66.1%	–	–
2021	66.2%	–	–
2022	67.9%	31.6%	3.1%
2023	72.2%	27.3%	3.9%
2024	70.6%	28.8%	9.3%
2025	66.2%	33.6%	4.9%

The play-by-play shotgun share, which has no source break, peaks at about 72% in 2023 and falls to about 66% in 2025. FTN's under-center share rises from 27% to 34% over the same seasons. Two independent recorders, one trend: quarterbacks really did move back under center.¹⁰ (The two columns add up to about 100% because the play-by-play flag counts the pistol as shotgun.)

They do not always agree on single plays, though, and Walker's run is one of the exceptions:

¹⁰Course calculation from nflverse play-by-play (`shotgun`) and FTN charting (`qb_location`), regular-season runs and dropbacks with a down, 2016–2025. The same trend in other sources: NFL Next Gen Stats reported offenses using fewer than three wide receivers on 41.7% of 2025 plays (James Reber, NFL.com, “NFL trend watch: heavy personnel could shape Super Bowl LX showdown between Patriots and Seahawks”): <https://www.nfl.com/news/nfl-trend-watch-heavy-personnel-could-shape-super-bowl-lx-showdown-between-patriots-and-seahawks>.

```

wf = ftn[(ftn.nflverse_game_id == "2025_22_SEA_NE") & (ftn.nflverse_play_id == 1015)].iloc[0]
wp_ = part[(part.nflverse_game_id == "2025_22_SEA_NE") & (part.play_id == 1015)].iloc[0]
print(f"pbp description:           {walker.desc[:30]}...   shotgun flag = {walker.shotgun:.0f}")
print(f"participation formation: {wp_.offense_formation} personnel: {wp_.offense_personnel}")
print(f"FTN charting:               qb_location = {wf.qb_location}, is_rpo = {wf.is_rpo}, "
      f"defenders in box (participation) = {wp_.defenders_in_box:.0f}")
both = fj[fj.season == 2025]
n_dis = int(((both.shotgun == 1) & (both.qb_location == "U")).sum())
print(f"2025 runs and dropbacks where pbp says shotgun but FTN says under center: {n_dis}")

```

```

pbp description:           (14:04) (Shotgun) 9-K.Walker l...   shotgun flag = 1
participation formation: UNDER CENTER personnel: 1 C, 2 G, 1 QB, 1 RB, 2 T, 1 TE, 3 WR
FTN charting:             qb_location = U, is_rpo = True, defenders in box (participation) = 8
2025 runs and dropbacks where pbp says shotgun but FTN says under center: 43

```

The statisticians typed “(Shotgun)”; FTN’s charter, watching film later, marked the quarterback under center and called the play a run-pass option. Without the film you can’t say who is right. It happens on a few dozen plays a season, too few to move a trend, and it is a useful reminder: every column in these tables is somebody’s judgement, recorded at speed.

Reproducing a chart end to end: personnel by season

Now put it together. Personnel showed how the league’s mix of personnel groupings moved from 2016 to 2025. Here is that chart rebuilt from nothing but the cells above: the `plays` funnel, the participation join, and a parser that reads both personnel formats.

```

def count(s, positions):
    """Sum the counts for some positions in a personnel string, in either format:
    2016-2022 (NGS) '1 RB, 2 TE, 2 WR'; 2023-2025 (FTN) '1 C, 2 G, 1 QB, 1 RB, 2 T, 2 TE, 2 WR'."""
    if not isinstance(s, str):
        return np.nan
    return sum(int(n) for n, pos in re.findall(r"(\d+)\s+([A-Z]+)", s) if pos in positions)

pers = joined[["season", "offense_personnel"]].copy()
pers["rb"] = pers.offense_personnel.map(lambda s: count(s, {"RB", "FB"})) # fullbacks are backs
pers["te"] = pers.offense_personnel.map(lambda s: count(s, {"TE"}))
pers = pers.dropna(subset=["rb", "te"])
pers["pers"] = pers.rb.astype(int).astype(str) + pers.te.astype(int).astype(str)
share = pd.crosstab(pers.season, pers.pers, normalize="index") * 100
md_table(share[["11", "12", "13", "21"]].round(1).reset_index(), digits=1)

```

season	11	12	13	21
2016	60.7	18.8	3.1	7.9
2017	59.6	20.6	4.2	8.4
2018	64.6	17.8	3.5	8.0
2019	60.3	20.9	3.0	7.9
2020	59.4	20.1	3.6	8.3
2021	60.5	21.6	3.6	7.3
2022	62.8	20.1	4.1	8.2
2023	64.2	20.4	3.4	7.4
2024	62.7	23.5	3.7	6.3

season	11	12	13	21
2025	58.8	24.4	5.1	7.0

```

lines = {"11": style.OFFENSE, "12": style.DEFENSE, "21": style.THIRD, "13": "#eda100"}
other = 100 - share[list(lines)].sum(axis=1)
ends = {k: share.loc[2025, k] for k in lines}
ends["all other"] = other.loc[2025]
pos, last = {}, -99.0 # nudge end labels apart so close values don't collide
for k, v in sorted(ends.items(), key=lambda kv: kv[1]):
    pos[k] = max(v, last + 3.6)
    last = pos[k]
fig, ax = plt.subplots(figsize=(6.5, 3.1))
for k, col in lines.items():
    ax.plot(share.index, share[k], color=col, lw=2.2, marker="o", ms=3.5)
ax.plot(other.index, other, color=style.MUTED, lw=1.4, ls=(0, (3, 2)))
for k, v in ends.items():
    ax.text(2025.3, pos[k], f"{k}: {v:.1f}%", va="center", fontsize=8, fontweight="bold",
           color=lines.get(k, style.INK_2))
ax.axvline(2022.5, color=style.BASELINE, lw=1, ls=(0, (1, 2)))
ax.text(2022.55, 46, "source change\n(NGS → FTN)", fontsize=7, color=style.INK_2, va="center")
ax.set_xlim(2015.6, 2027.6)
ax.set_xticks(range(2016, 2026), [str(y) for y in range(2016, 2026)], fontsize=8)
ax.set_ylim(0, 70)
ax.set_ylabel("share of plays (%)", fontsize=8)
ax.set_title("Personnel share by season (rebuilt)", fontsize=10)
plt.show()

```

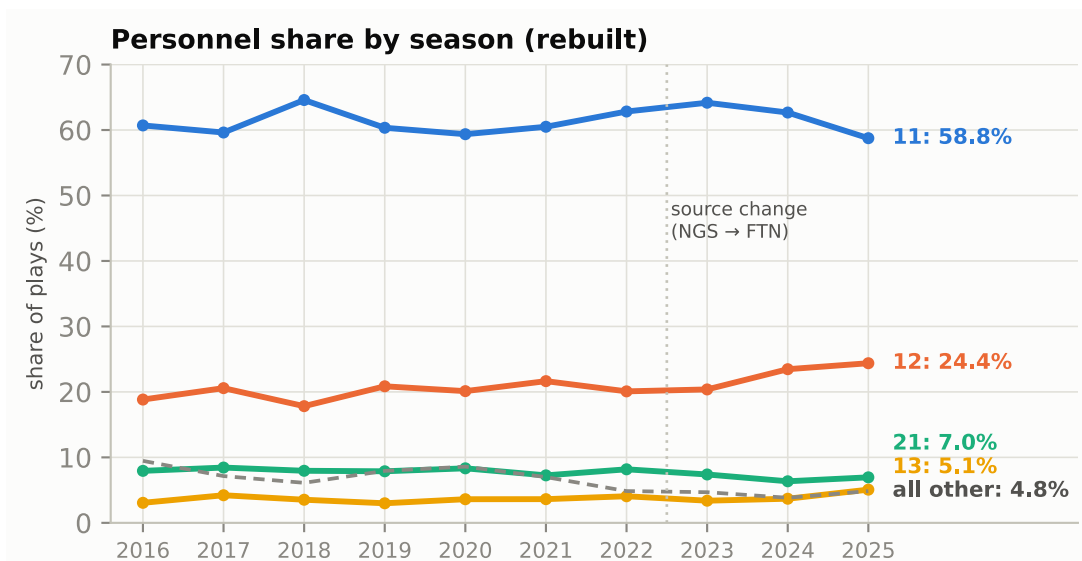


Figure 8: The personnel chart from 02-01, rebuilt end to end in this chapter: share of NFL regular-season runs and dropbacks by offensive personnel, 2016–2025 (nflverse play-by-play joined to participation data; backs = RB + FB, then tight ends). 11 personnel held near 60% for a decade and fell to 58.8% in 2025; 12 personnel rose to 24.4%. The dotted line marks the change of supplier; personnel is one of the fields that survives it, because both suppliers count backs and tight ends.

Same numbers as 02-01, to the decimal: 58.8% for 11 personnel in 2025, 24.4% for 12, 5.1% for 13. That is the point of the exercise. A chart you can rebuild is a chart you can extend: change SEASONS, filter joined to one team (joined.posteam == "SEA"), or split by down, and the same dozen lines answer a new question.

When you reproduce someone else's chart, three things usually explain a mismatch, in this order: **the filter** (regular season only? penalties? scrambles?), **the counting rule** (here, does a fullback count as a back? the course says yes), and **the data vintage**

(nflverse occasionally re-releases past seasons; FTN re-released 2023 and 2024 participation in September 2025). Check them before you suspect the arithmetic.

Film room: league pass rate by down, reproduced

The first chart in this book that predicted anything was Downs, Distance, and Field Position's grid of how often offenses passed by down and distance in 2025. Here it is again, rebuilt with that chapter's filter (win probability 20% to 80%, outside the last two minutes of each half) and its buckets:

```
n25 = snaps[(snaps.season == 2025) & (snaps.half_seconds_remaining > 120)
            & snaps.wp.between(0.2, 0.8) & (snaps.down <= 3)].copy()
BUCKETS = ["1", "2-3", "4-6", "7-10", "11+"]
n25["dist"] = pd.cut(n25.ydstogo, [0, 1, 3, 6, 10, 99], labels=BUCKETS)
grid = n25.groupby(["down", "dist"], observed=False)["pass"].agg(["mean", "size"]).reset_index()
cmap = LinearSegmentedColormap.from_list("pr", [style.DEFENSE_TINT, "#f6f4ee", style.OFFENSE_TINT,
                                                style.OFFENSE])

fig, ax = plt.subplots(figsize=(6.5, 2.6))
for _, r in grid.iterrows():
    i, j = int(r.down) - 1, BUCKETS.index(str(r.dist))
    rare = r["size"] < 250
    dark = (not rare) and r["mean"] >= 0.8
    ax.add_patch(Rectangle((j, 2 - i), 0.96, 0.92, ec="white", lw=1.5,
                           fc=style.GRID if rare else cmap(float(r["mean"]))))
    ax.text(j + 0.48, 2 - i + 0.55, "rare" if rare else f"{r['mean']:.0%}", ha="center",
            va="center", fontsize=8 if rare else 10.5, fontweight="bold",
            color="white" if dark else style.INK)
    if not rare:
        ax.text(j + 0.48, 2 - i + 0.24, f"{int(r['size']):,} plays", ha="center", va="center",
                fontsize=6.3, color="white" if dark else style.INK_2)
ax.set_xlim(-0.05, 5)
ax.set_ylim(-0.05, 3)
ax.set_xticks([k + 0.48 for k in range(5)], ["1 yd", "2-3 yds", "4-6 yds", "7-10 yds",
                                             "11+ yds"], fontsize=7.5)
ax.set_yticks([2.46, 1.46, 0.46], ["1st down", "2nd down", "3rd down"], fontsize=8)
ax.tick_params(length=0)
ax.grid(False)
for sp in ax.spines.values():
    sp.set_visible(False)
ax.xaxis.tick_top()
plt.show()
```

	1 yd	2–3 yds	4–6 yds	7–10 yds	11+ yds
1st down	rare	rare	rare	46% 7,445 plays	69% 318 plays
2nd down	26% 344 plays	32% 645 plays	48% 1,543 plays	69% 2,710 plays	82% 891 plays
3rd down	21% 487 plays	64% 678 plays	93% 914 plays	97% 1,072 plays	92% 681 plays

Figure 9: 01-02’s pass-rate grid, rebuilt: how often the offense passed, by down and distance, 2025 regular season, neutral situations as 01-02 defined them (outside the last two minutes of each half, win probability 20–80%). A pass is any dropback, so sacks and scrambles count. Cells with fewer than 250 plays are marked ‘rare’. Source: nflverse play-by-play.

Same grid, same numbers. Now do what 01-02 couldn’t, because it had no code: ask whether the pattern is new. Play-by-play goes back to 1999, so the same question can be asked of 27 seasons. This time the cell uses the course’s garbage-time filter, and it tracks the first three downs (fourth down mixes “how often do they pass?” with “how often do they go for it at all?”, which is Win Probability and Decision Analysis’s subject):

```
gt = snaps[snaps.wp.between(0.10, 0.90) & (snaps.half_seconds_remaining > 120)]
by = gt.groupby(["season", "down"])[["pass"].mean().unstack() * 100
DCOL = {1: style.OFFENSE, 2: style.THIRD, 3: style.DEFENSE}
fig, ax = plt.subplots(figsize=(6.5, 3.0))
for d_, col in DCOL.items():
    ax.plot(by.index, by[d_], color=col, lw=2.0, marker="o", ms=2.6)
    ax.text(2025.6, by.loc[2025, d_], f"{'1st', '2nd', '3rd'}[d_] down: "
           f"{by.loc[2025, d_]:.0f}%", va="center", fontsize=8, fontweight="bold", color=col)
ax.set_xlim(1998.5, 2030.5)
ax.set_xticks(range(1999, 2026, 2), [f"{y}" if y % 4 == 3 else "" for y in range(1999, 2026, 2)],
              fontsize=7.5)
ax.set_ylim(35, 95)
ax.set_ylabel("pass rate (%)", fontsize=8)
ax.set_title("Pass rate by down, 1999–2025", fontsize=10)
plt.show()
print(by.loc[[1999, 2009, 2019, 2025], [1, 2, 3, 4]].round(1).to_string())
```

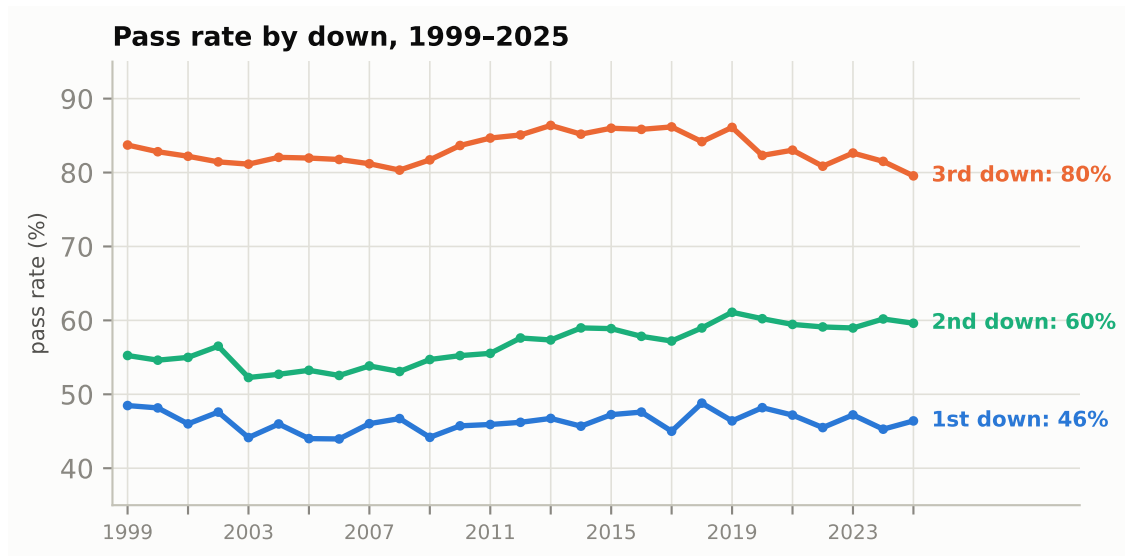


Figure 10: League pass rate by down, every regular season from 1999 to 2025, with the course’s garbage-time filter (win probability 10–90%, outside the last two minutes of each half); a pass is any dropback, including sacks and scrambles. First down has hardly moved in 27 seasons. Second down drifted up by about five points. Third down peaked around 2013–2019 and has eased since.

Source: nflverse play-by-play.

down	1.0	2.0	3.0	4.0
season				
1999	48.5	55.2	83.7	42.7
2009	44.2	54.7	81.7	39.7
2019	46.4	61.1	86.1	54.5
2025	46.4	59.6	79.5	56.8

i Film room: the league, 1999–2025

What to notice in a chart you built yourself:

- **First down is the most stable number in football.** In every season from 1999 to 2025, with the game still in doubt, offenses passed on between 44% and 49% of first downs. Twenty-seven seasons of rule changes that favoured the passing game (the illegal-contact emphasis of 2004, the protection of defenseless receivers and quarterbacks) and the spread of shotgun offenses barely moved it. First-and-10 is where play-callers most want to be unpredictable, and they are.
- **The passing boom showed up on second down.** Second-down pass rate climbed from about 55% to about 60%. Coaches who once ran on second down to “set up third-and-manageable” increasingly throw to convert on second down instead.
- **Third down eased after the late 2010s.** Third-down pass rates peaked around 86% in the 2013–2019 seasons and fell below 80% in 2025, consistent with more third-and-short runs, more under-center play and more quarterback sneaks and pushes. A chart can’t tell you *why*; the chapters on third down and the modern era argue the causes.
- **Fourth down rose most of all** (see the printed table): about 43% passes on fourth-down runs and dropbacks in 1999 against about 57% in 2025, consistent with teams going for it more often at longer distances, where a pass is the natural call.

The lesson for reading a game is the same one the 01-02 grid taught: before personnel, before formation, first-and-10 is a coin flip and third-and-long is not. The lesson for working with data is new: the 2025 grid was one season’s snapshot, and a few lines of code turned it into a 27-season history, which is where most good questions start.

From a row to a diagram: nflverse meets gridiron

The book’s diagram library, `gridiron`, and `nflverse` speak almost the same language. A row gives you the situation (`yardline_100`, `ydstogo`), and participation and FTN add the personnel, the formation and the box count. What no row gives you is **where each player stood**: that needs tracking data, the subject of Tracking Data and the Big Data Bowl. So a diagram built from a row is part data and part illustration, and the caption has to say which part is which. Here is Walker’s run, built from its row:

```

los = 110 - float(walker.yardline_100)
off = offense("singleback") # 11 personnel, QB under center (FTN: U)
dfn = defense("nickel", "single_high", off) # 3 CB; NB and SS make 8 in the box
grouping = (f"{count(wp_.offense_personnel, {'RB', 'FB'})}"
            f"{count(wp_.offense_personnel, {'TE'})} personnel")
row_play = Play(off, dfn, los=los, to_go=float(walker.ydstogo),
               title=f"{'', '1st', '2nd', '3rd', '4th'}[int(walker.down)] & {walker.ydstogo:.0f} at SEA "
                   f"{110 - walker.yardline_100 - 10:.0f} · {grouping} (from the data)")
row_play.handoff("RB", t=0.8)
row_play.run("RB", [(3.0, -3.5), (7.0, -8.5), (17.0, -12.0)])
row_play.highlight("RB")
row_play.note(f"+{walker.yards_gained:.0f} yds to NE 46", at=(13.0, -10.5))
fld = row_play.draw(window=(-9, 15), lateral=(-20, 20))
drop_numbers(fld.ax)
corners = [row_play.to_abs(d_, w_) for d_, w_ in [(0.1, -6.0), (8.2, -6.0), (8.2, 7.6), (0.1, 7.6)]]
bx, by = fld.xy(np.array([c[0] for c in corners]), np.array([c[1] for c in corners]))
fld.ax.add_patch(plt.Polygon(np.c_[bx, by], closed=True, fill=False, ec=style.DEFENSE_DARK,
                             lw=1.0, ls=(0, (3, 2)), zorder=3))
tx, ty = fld.xy(*row_play.to_abs(9.1, 4.6))
fld.ax.text(tx, ty, "the box: 8 defenders", fontsize=7, color=style.DEFENSE_DARK, ha="center",
            va="center", zorder=7, bbox=dict(boxstyle="round,pad=0.2", fc=style.SURFACE,
            ec="none"))
gx, gy = fld.xy(*row_play.to_abs(-5.0, 0.0))
fld.ax.add_patch(Circle((gx, gy), 0.9, fill=False, ec=style.OFFENSE_DARK, lw=1.0, ls=(0, (2, 2)),
                        zorder=7))
fld.ax.text(gx + 1.6, gy, "pbp: '(Shotgun)'", fontsize=6.8, color=style.OFFENSE_DARK,
            va="center", zorder=7)
plt.show()

```

2nd & 10 at SEA 24 · 11 personnel (from the data)

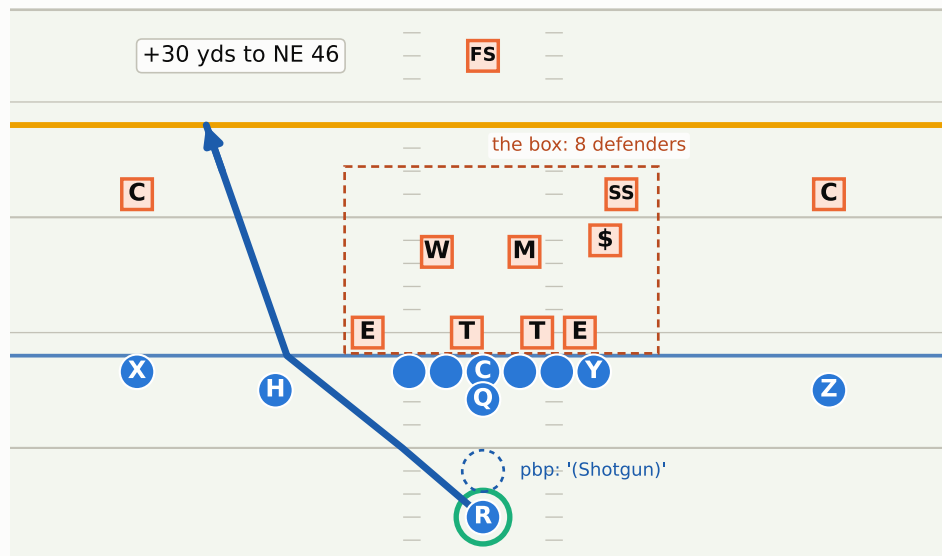


Figure 11: Walker's run drawn from its data. From the row: second-and-10, ball at yardline₁₀₀ = 76 (so the line of scrimmage is at BDB x = 34 and the yellow line 10 yards on), a run to the left end for 30 yards. From participation: 11 personnel (one back, one tight end, three receivers) against three cornerbacks (a nickel defense) with eight defenders in the box (dashed outline). From FTN: quarterback under center. Everything else (each player's exact spot, the blocking) is the course's typical alignment, not the real one. The dashed ring is where the play description's '(Shotgun)' would put the quarterback.

The title is built from the data, so the same cell draws any row you point it at. The picture also produces **simulated tracking data**: positions ten times a second, in exactly the column layout the Big Data Bowl uses. That is how this book animates hand-drawn plays, and it means the code that animates them will animate real tracking frames unchanged.

```
trk = row_play.to_tracking(game_id=2026020800, play_id=1015) # simulated, BDB-format
print(f"{len(trk):,} rows: {trk.frameId.nunique()} frames x {trk.displayName.nunique()} "
      f"tracks (22 players + the ball)")
rb = trk[trk.displayName == "RB"].iloc[[0, 5, 8, 12, 20, 28]]
md_table(pd.DataFrame({"frameId": rb.frameId, "frameType": rb.frameType,
                       "time": rb.time.map(lambda v: f"{v + 0.0:+.1f}"),
                       "x": rb.x.round(1), "y": rb.y.round(1), "s": rb.s.round(1),
                       "dir": rb.dir.map(lambda v: "-" if pd.isna(v) else f"{v:.0f}"),
                       "event": rb.event.fillna("")})), digits=1)
```

1,035 rows: 45 frames x 23 tracks (22 players + the ball)

frameId	frameType	time	x	y	s	dir	event
1	BEFORE_SNAP	-0.5	27.0	26.7	0.0	-	
6	SNAP	+0.0	27.0	26.7	3.3	41	ball_snap
9	AFTER_SNAP	+0.3	28.3	28.1	6.5	41	
13	AFTER_SNAP	+0.7	30.0	30.1	6.5	40	
21	AFTER_SNAP	+1.5	33.2	34.2	6.5	39	
29	AFTER_SNAP	+2.3	37.7	36.5	6.5	71	

```
show_animation(row_play, times=[0.0, 0.8, 1.6, 2.6], title="Simulated, BDB-format")
```

Simulated, BDB-format

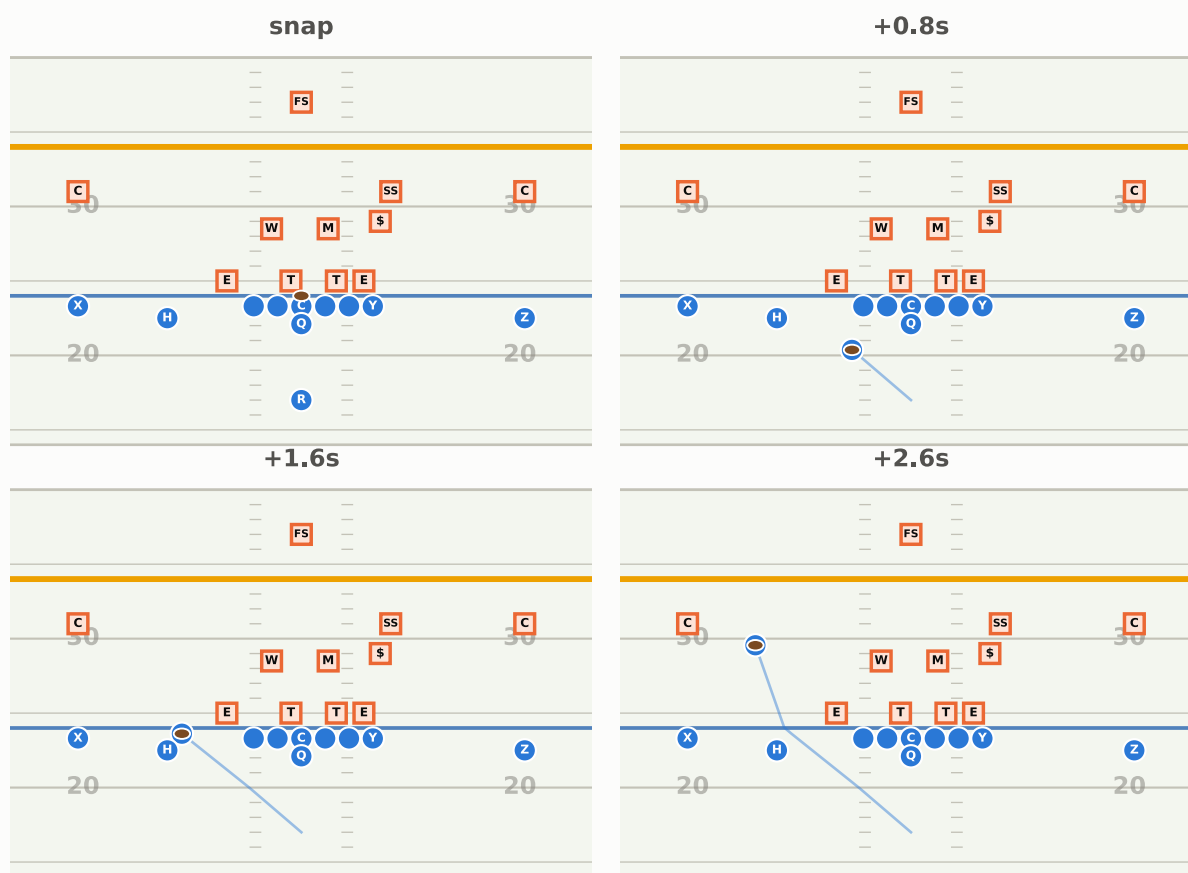


Figure 12: Walker's run as simulated, BDB-format tracking (generated from the diagram above, not real player positions): the snap, the handoff, the back bending to the left end, and the run past the yellow line. In the web version this is an animation; in print, a frame strip.

With real Big Data Bowl files the drawing code is the same; only the source of the frames changes. The cell below is not run in this book (the Kaggle files are not redistributed), but it is what you will write in 13-05:

```
import pandas as pd
from gridiron import bdb, show_animation

trk = bdb.load_tracking("data/bdb/tracking_week_1.csv")    # BDB 2025: the 2022 season
plays_bdb = pd.read_csv("data/bdb/plays.csv")
one = bdb.prepare(trk, plays_bdb, game_id=2022091100, play_id=55)  # flips plays moving left
show_animation(one, times=[-1.0, 0.0, 1.5, 3.0])
# BDB plays.csv also carries nflverse-style keys, so the pbp row (EPA, WP) joins on
# old_game_id == gameId and play_id == playId.
```

Watch for it on Sunday: the Sunday notebook

You have kept a prediction log since How to Watch a Broadcast: before each snap, a run or pass call with a confidence and the clue you used. Scoring it by hand means writing down every result as well. With nflverse you don't have to: the data fills in the results the next morning.

During the season, nflverse refreshes the current season's play-by-play as games finish, so your weekly routine looks like this (not run here, because the book is built from completed seasons):

```
import nflreadpy as nfl
from nflreadpy.config import update_config

update_config(cache_mode="filesystem", cache_duration=6 * 60 * 60) # refresh after 6 hours
season = nfl.get_current_season()
week_pbp = nfl.load_pbp([season]).to_pandas()
week_pbp = week_pbp[week_pbp.week == week_pbp.week.max()] # the latest week in the file
```

Then join your log to the plays and score it. Here is a worked example on Super Bowl LX, with an example log of Seattle's first fifteen snaps, written the way you would write it from the couch: the clock as you read it off the scorebug (often a few seconds off), the down and distance, your call and your confidence. The join can't rely on an exact clock, so it matches each line to the play with the same quarter, down and distance and the **nearest** clock reading, within 15 seconds:

```
log = pd.DataFrame(
    [(1, "15:00", 1, 10, "R", .60, "sit"), (1, "14:31", 1, 10, "P", .55, "align"),
     (1, "13:52", 2, 10, "P", .65, "sit"), (1, "13:27", 1, 10, "R", .55, "sit"),
     (1, "13:00", 1, 10, "P", .60, "align"), (1, "12:53", 2, 10, "R", .55, "align"),
     (1, "12:08", 3, 7, "P", .85, "sit"), (1, "07:13", 1, 10, "R", .60, "sit"),
     (1, "07:08", 2, 10, "P", .65, "align"), (1, "06:24", 3, 6, "P", .85, "sit"),
     (1, "03:31", 1, 10, "R", .65, "align"), (1, "02:53", 2, 8, "R", .55, "align"),
     (1, "02:09", 3, 7, "P", .85, "sit"), (2, "14:12", 1, 10, "R", .55, "sit"),
     (2, "14:06", 2, 10, "P", .60, "sit")],
    columns=["qtr", "clock", "down", "togo", "call", "conf", "clue"])

def secs(mmss):
    m, s = mmss.split(":")
    return int(m) * 60 + int(s)

game = df25[(df25.game_id == "2025_22_SEA_NE") & (df25.posteam == "SEA")
             & df25.play_type.isin(["pass", "run"]) & df25.down.notna()].copy()
game["t"] = game.time.map(secs)
game["result"] = np.where(game["pass"] == 1, "P", "R") # sacks and scrambles are passes
log["t"] = log.clock.map(secs)
scored = pd.merge_asof(
    log.astype({"qtr": float, "down": float, "togo": float}).sort_values("t"),
    game.rename(columns={"ydstogo": "togo"}).sort_values("t")[["qtr", "down", "togo", "t",
                                                                "result", "epa", "desc"]],
    on="t", by=["qtr", "down", "togo"], direction="nearest", tolerance=15)
scored = scored.sort_values(["qtr", "t"], ascending=[True, False])
scored["hit"] = scored.call == scored.result
# The down-and-distance guesser from 01-06: run on short 1st/2nd/4th downs, pass otherwise
dd_run = ((scored.togo <= 3) & (scored.down != 3)) | ((scored.down == 1) & (scored.togo <= 6))
scored["dd_hit"] = scored.result == np.where(dd_run, "R", "P")
print(f"matched {scored.result.notna().sum()} of {len(scored)} log lines")
print(f"You: {scored.hit.mean():.0%} always pass: {(scored.result == 'P').mean():.0%} "
      f"down & distance only: {scored.dd_hit.mean():.0%}")
md_table(scored[["qtr", "clock", "down", "togo", "call", "result", "hit", "clue"]].
          .astype({"down": int, "togo": int, "hit": str}))
```

```
matched 15 of 15 log lines
You: 67% always pass: 60% down & distance only: 60%
```

qtr	clock	down	togo	call	result	hit	clue
1	15:00	1	10	R	R	True	sit
1	14:31	1	10	P	R	False	align
1	13:52	2	10	P	P	True	sit
1	13:27	1	10	R	P	False	sit
1	13:00	1	10	P	P	True	align
1	12:53	2	10	R	R	True	align
1	12:08	3	7	P	P	True	sit
1	07:13	1	10	R	P	False	sit
1	07:08	2	10	P	P	True	align
1	06:24	3	6	P	P	True	sit
1	03:31	1	10	R	R	True	align
1	02:53	2	8	R	R	True	align
1	02:09	3	7	P	P	True	sit
2	14:12	1	10	R	P	False	sit
2	14:06	2	10	P	R	False	sit

Ten of fifteen, against nine for the down-and-distance guesser and nine for “always pass.” Fifteen snaps prove nothing, of course; Expected Points, EPA, and Success Rate shows how many plays it takes before a rate means anything, and Film Study: All-22, Charting a Game, and Scoring Yourself scores a season’s log properly, with the confidence column. But the habit is the thing: the log you keep on Sunday is a table, and the results arrive as a table on Monday.

💡 Watch for it: the Sunday notebook

- **During the game:** keep the log exactly as in 01-06 (quarter, clock, down, distance, call, confidence, clue). Write the clock as the scorebug shows it just before the snap; being a few seconds off is fine, because the join looks for the nearest play.
- **Monday morning:** load the newest week of play-by-play with a short cache, filter to the game and the offense you watched, and run the join above. Check the match count first: a log line that finds no play usually has a mistyped down or distance.
- **Score it three ways:** your hit rate, “always pass,” and the down-and-distance guesser. Beat both, and you are reading something the scorebug doesn’t show.
- **Read your misses.** Sort the misses by your confidence and read their `desc` text. Confident misses on first-and-10 usually mean you missed a clue in the formation; misses on third down usually mean a draw or a screen you didn’t see coming.
- **Keep the season in one file.** Append each week’s scored log to a CSV. By midseason you can group by `clue` and see which of your reads actually work.

Predict the play

These drills ask you to predict the *data* as well as the football: what the row will say, and what your filters will do with it. The diagrams use the course’s position letters: on offense Q is the quarterback, R the running back, X, Z and H wide receivers, Y and U tight ends; on defense E is an end, T a tackle, S, M and W the linebackers, C a cornerback, \$ the nickel back, and FS and SS the safeties.

💡 Predict the play 1: write the row

Third-and-2 at the opponent's 40, second quarter. The offense brings two tight ends and lines up under center; the defense answers with its base 4-3 and one deep safety. The back takes the handoff and gains **3 yards**. Before you look: what will this play's row say for `play_type`, `pass`, `rush`, `down`, `ydstogo`, `yardline_100`, `yards_gained` and `first_down`? And what will the *next* row's `down`, `ydstogo` and `yardline_100` be?

```
off = offense("ace")
p = Play(off, defense("4-3_over", "single_high", off), los=70, to_go=2,
        title="3rd & 2 at the opponent's 40")
p.highlight("RB")
p.highlight("Y", "U", color=style.LINE_TO_GAIN)
drop_numbers(p.draw(lateral=(-19, 19), window=(-9, 15)).ax)
plt.show()
```

3rd & 2 at the opponent's 40

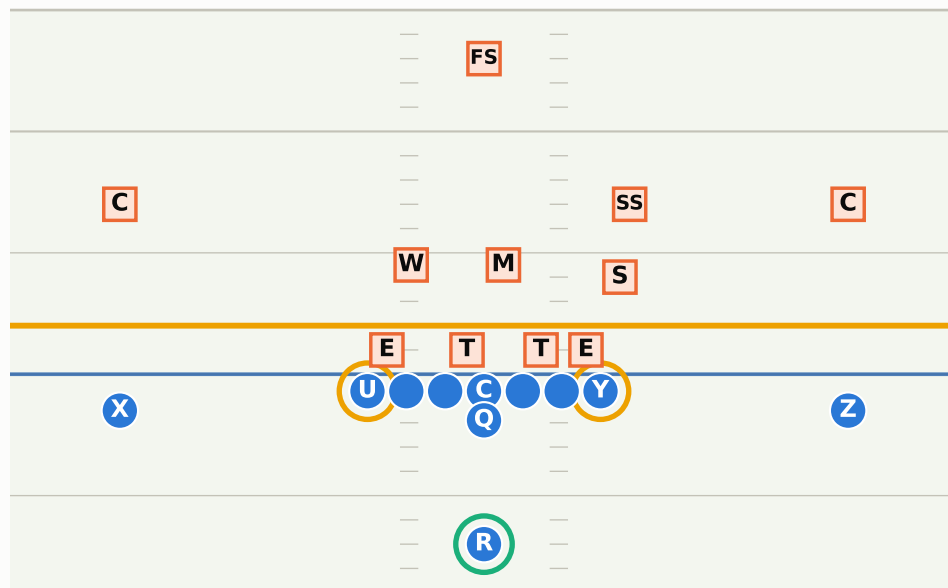


Figure 13: Drill 1: third-and-2 at the opponent's 40. 12 personnel (tight ends Y and U, gold rings) in an under-center 'ace' set against a base 4-3 with one deep safety. The running back (R, aqua ring) will carry for 3 yards. The yellow line is the line to gain, 2 yards ahead.

Answer 1 is at the end of the chapter. Try it first.

💡 Predict the play 2: the scramble

Third-and-8 at your own 35. The offense is in the shotgun with three receivers; the defense plays two deep safeties with everyone covered. The quarterback drops, finds nothing, escapes to his right and runs for **9 yards** and a first down. How will nflverse file this play? Will it count as a pass under `play_type == "pass"`? Under `pass == 1`? Which should a pass-rate chart use, and how much does the choice matter over a season?

```
off = [q.moved(w=-2.3) if q.key == "RB" else q for q in offense("gun_2x2")]
p = Play(off, defense("nickel", "two_high", off), los=45, to_go=8,
        title="3rd & 8 at your own 35")
p.route("X", [(0, 0), (10, 0)])
p.route("Z", [(0, 0), (10, 0)])
p.route("H", [(0, 0), (6, -3)])
p.route("Y", [(0, 0), (6, -3)])
p.path("QB", [(-2.0, 0.0), (-1.5, 4.5), (14.0, 8.0)])
p.highlight("QB")
p.note("scramble: +9", at=(10.8, 3.0))
drop_numbers(p.draw(lateral=(-19, 19), window=(-9, 15)).ax)
plt.show()
```

3rd & 8 at your own 35

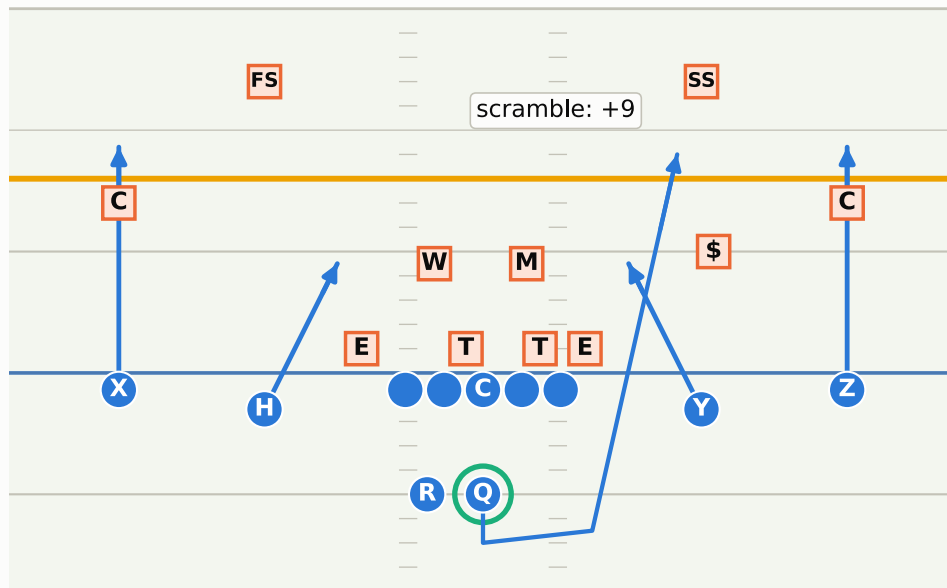


Figure 14: Drill 2: third-and-8 at the offense's own 35. Shotgun 11 personnel against a nickel defense with two deep safeties. The receivers' routes are covered, so the quarterback (Q, aqua ring) drops, slides right and runs past the yellow line for 9 yards.

Answer 2 is at the end of the chapter. Try it first.

💡 Predict the play 3: called back

First-and-10 at your own 25. The offense runs off right tackle for **12 yards**, but there is a flag at the left end of the line: offensive holding on the left tackle, accepted. In `nflverse`, what is this play's `play_type`, and will your “real plays” filter keep it? What sign will its EPA have? Would the `nflfastR` guide's filter, `pass == 1 | rush == 1`, keep it?

```
off = offense("singleback")
p = Play(off, defense("4-3_over", "single_high", off), los=35, to_go=10,
         title="1st & 10 at your own 25")
p.handoff("RB", t=0.8)
p.run("RB", [(5.0, 2.5), (8.0, 3.6), (19.0, 4.5)])
p.block("LT", target="WDE")
p.highlight("RB")
p.highlight("LT", color=style.DEFENSE)
p.note("+12 yards", at=(12.0, 9.5))
p.note("flag: holding (LT)", at=(-4.5, -9.5))
drop_numbers(p.draw(lateral=(-19, 19), window=(-9, 16)).ax)
plt.show()
```

1st & 10 at your own 25

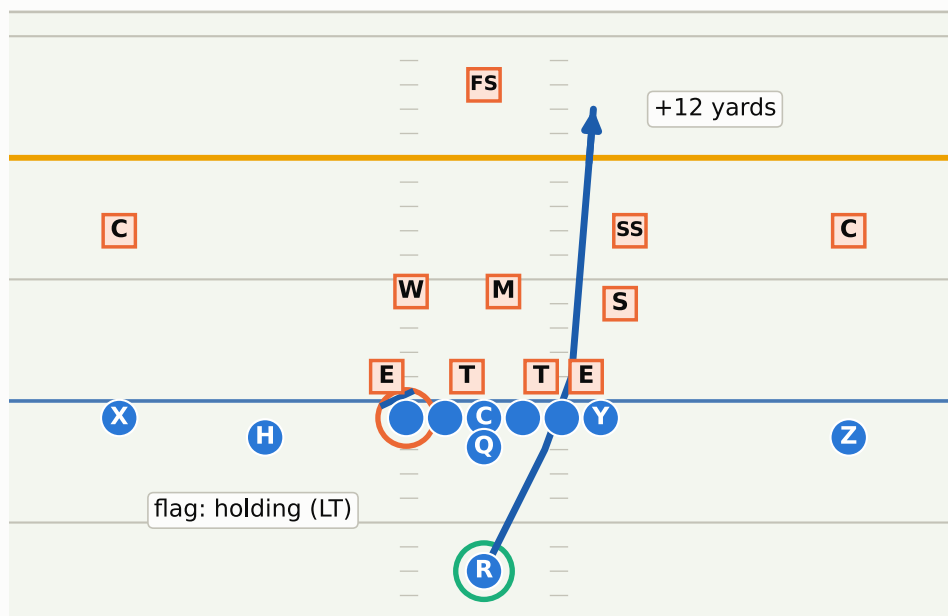


Figure 15: Drill 3: first-and-10 at the offense's own 25. An under-center 11-personnel set against a base 4-3. The running back (R, aqua ring) runs off right tackle for 12 yards, but the left tackle (orange ring) is flagged for holding the defensive end, away from the play.

Answer 3 is at the end of the chapter. Try it first.

Takeaways

- **nflverse is free, public, reproducible football data:** play-by-play for every play since 1999, plus schedules, rosters, participation and FTN charting, with `nflfastR`'s models (EP, EPA, WP, CPOE, xpass) on every row. In Python you load it with **`nflreadpy`** (Polars; `.to_pandas()` when you want pandas), not the archived `nfl_data_py`.
- **A row is a scorebug plus a result plus model estimates**, and not every row is a play. Situation columns (`down`, `ydstogo`, `yardline_100`) describe the moment before the snap; `yardline_100` is the yards still to go for a touchdown, which is why it falls as an offense advances.

- **Count passes with `pass == 1`**, which includes sacks and scrambles, and decide on purpose what to do with no-play penalties, kneel-downs, two-point tries and garbage time. This book’s garbage-time filter keeps win probability 10–90% outside the last two minutes of each half. State your filter every time.
- **Participation and FTN charting add who was on the field and how they lined up**, but participation comes from two suppliers, with a source break between 2022 and 2023: the formation labels, personnel strings and coverage calls all change. Mark the break, translate to what both sides can say, and cross-check against a source that doesn’t break.
- **Reproduce before you extend**. Rebuilding 02-01’s personnel chart and 01-02’s pass grid to the decimal proves the pipeline; the same dozen lines then answer new questions, such as 27 seasons of pass rate by down.
- **Data can’t tell you where players stood**. A diagram built from a row is part data, part illustration, and the caption must say which; real positions need tracking data.

Connections

Builds on: Downs, Distance, and Field Position (down, distance, field position and the EPA primer, whose pass-rate grid we rebuilt) and Personnel (personnel groupings and the season-by-season chart). The prediction log comes from How to Watch a Broadcast; the formation vocabulary (shotgun, under center, pistol) from Formation Rules and Vocabulary.

Sets up:

- Expected Points, EPA, and Success Rate: builds the `ep` and `epa` columns from scratch with this chapter’s data and filters.
- Win Probability and Decision Analysis: the `wp` column, and the fourth-down decisions behind the chart above.
- Tendencies: PROE, Personnel, Motion, and Coverage: `xpass` and `pass_oe`, neutral situations, and coverage charts that respect the source break.
- Tracking Data and the Big Data Bowl: the real positions that a play-by-play row can’t give you, loaded with `gridiron.bdb`.
- Film Study: All-22, Charting a Game, and Scoring Yourself: the season’s prediction log, scored properly.

If you want to go deeper: the `nflreadr` data dictionaries for play-by-play, participation and FTN charting (the column definitions, straight from the source); `nflfastR`’s “A beginner’s guide to `nflfastR`,” written in R but easy to translate, for the standard filters; Ben Baldwin’s Open Source Football posts for how the models are built; and the `nflverse` Discord and GitHub issues, where the maintainers answer questions about data quirks.

i Go deeper: a checklist for any new `nflverse` question

1. **Which table, which seasons?** Check the timeline above before you promise a 2016–2025 chart of something that starts in 2018.
2. **Which rows?** Regular season? `pass == 1` or `play_type`? Penalty snaps? Two-point tries? Garbage time?
3. **Which key?** `game_id + play_id` for participation; `nflverse_game_id + nflverse_play_id` for FTN; `gsis_id` for players. Print the match rate after every join.
4. **Does a known total come out right?** A player’s yards, a team’s record, a published league rate.
5. **Does the series cross a source break?** Mark it, translate it, or cross-check it.
6. **Say all of the above in the caption.**

Answers to Predict the play

Answer 1.

The row records the situation **before** the snap and the result in separate columns: `play_type = run`, `pass = 0`, `rush = 1`, `down = 3`, `ydstogo = 2`, `yardline_100 = 40`, `yards_gained = 3`, `first_down = 1`. The next row is a new series: `down = 1`, `ydstogo = 10`, `yardline_100 = 37`, because the ball moved 3 yards *closer* to the goal line and `yardline_100` counts down as an offense advances. The EPA is positive: a third down became a fresh first down.

Two things trip people up. First, `yardline_100` falls as the offense gains ground, so “plays inside the 20” is `yardline_100 <= 20`, not `>= 80`. Second, the row’s situation is **before** the snap: to know where a play ended you read the next row, or compute `yardline_100 - yards_gained` (which is right for an ordinary run or completion and wrong after a turnover or a penalty, which is why `nflverse` also stores `end_yard_line`).

Answer 2.

nflverse files it as `play_type = run`, because the quarterback carried the ball, with `qb_scramble = 1`, `qb_dropback = 1`, `pass = 1` and `rush = 0`. So `play_type == "pass"` counts it as a run, and `pass == 1` counts it as a pass. A chart about **play-calling** should use `pass == 1`: the offense called a pass, and the defense defended one. Scrambles are not rare: there were 1,089 in the 2025 regular season, and the league's pass rate on runs and dropbacks was 60.1% counted by the flag against 56.8% counted by `play_type`. That is a gap of about three points, as big as the effect of a garbage-time filter.

(A chart about **who carried the ball**, such as a quarterback's rushing yards, is different: there the scramble yards are rushing yards, as the box score says.)

Answer 3.

The play never happened, officially. nflverse records it as `play_type = no_play` with `penalty_type = "Offensive Holding"`, `yards_gained = 0` and `rush = 1` (the run is still flagged). The course's filter, `play_type` in pass or run, **drops** it; `pass == 1 | rush == 1` **keeps** it. Its EPA is clearly **negative**: instead of first-and-10 at the 37, the offense faces first-and-20 at its own 15 (a 10-yard penalty from the line of scrimmage). In the 2025 regular season there were 176 holding calls that wiped out runs, and they cost -1.12 expected points each on average.

Which filter is right depends on the question. "How good is this team's running game?" arguably should charge the offense for its holds, which is why nflfastR's guide keeps penalty snaps for EPA. "How often does this team call a run?" is cleaner without them. Expected Points, EPA, and Success Rate measures the difference: leaving penalty snaps out makes passing look a little worse, because defensive pass interference is mostly a deep-ball outcome. Neither choice is wrong. Not knowing which one you made is.